

AWS Certified Solutions Architect Associate

Memorization Notes · SAA-C03

How to use this

Every page is built for recall, not for first-time learning. Cover the right-hand column of any table, say the answer out loud, then uncover and check.

Cue → **Answer** tables are facts and definitions. **Numbers** tables are pure rote — do those first. **Mnemonics** are for lists you must produce from nothing. **Triggers** map an exam scenario to the service that answers it, and are worth the most marks per minute of study.

Highlighted callouts are traps — the places where a plausible-looking answer is wrong.

Start with **Numbers and Limits**, finish with **Decision Triggers**, and drill **Rapid Fire** daily in the last two weeks.

AWS Certified Solutions Architect – Associate (SAA-C03)

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01 — Numbers and Limits

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AWS Certified Solutions Architect – Associate (SAA-C03)

Memorization Notes — Index

These notes are built for **recall under time pressure**, not for first-time learning. Every page uses one of four shapes:

Shape	What it looks like	Use it for
Cue → Answer	A two-column table. Cover the right column.	Facts and definitions
Numbers table	A limit, size, or duration per row	Hard numbers the exam asks directly
Mnemonic	A word or sentence that unpacks into a list	Lists you must produce from nothing
Trigger	"If the question says X → answer Y"	Scenario questions (most of the exam)

■ How to drill these

1. **Day 1–2:** Read `01-numbers-and-limits.md` and `02-mnemonics.md` out loud. These are pure rote. Nothing else on this list sticks until the numbers do.
2. **Day 3–7:** One service page per day (`03` through `09`). Cover the answer column, say the answer, uncover, check.
3. **Day 8–9:** `10-decision-triggers.md` . This page is worth the most marks per minute of study. The exam is mostly scenario questions and this page is the scenario-to-service map.
4. **Day 10 onward:** `11-rapid-fire.md` every morning. 150 one-liners, about 12 minutes. If you miss one, go back to its service page.
5. Read `99-review-of-original-notes.md` once, early. It lists the facts in the older slide decks that are **now wrong** — unlearning a stale number is harder than learning a new one, so do it before you drill.

■ Exam shape (SAA-C03)

Item	Value
Questions	65 (50 scored, 15 unscored)
Time	130 minutes
Passing score	720 / 1000
Question types	Multiple choice (1 of 4), multiple response (2+ of 5+)
Cost	150 USD

■ Scored domains — know the weights, they tell you where to spend study time

Domain	Weight	Mnemonic hook
1. Design Secure Architectures	30%	S ecure
2. Design Resilient Architectures	26%	R esilient

Domain	Weight	Mnemonic hook
3. Design High-Performing Architectures	24%	H igh-performing
4. Design Cost-Optimized Architectures	20%	C ost

Mnemonic: **"SRHC" — Security Rules, Hardware Costs.** Security is the biggest single slice. When two answers both work, the more secure one usually wins.

■ The six pillars of the Well-Architected Framework

Mnemonic: **"COPS RS"** — or remember the sentence **"Our Six Pillars Really Count, Seriously."**

Pillar	One-line test
O perational Excellence	Can we run and improve it?
S ecurity	Who can touch what, and is it logged?
R eliability	Does it survive a failure?
P erformance Efficiency	Is it the right-sized, right-shaped resource?
C ost Optimization	Are we paying for idle?
S ustainability	Are we wasting energy? (added 2021 — newer than the old slide decks)

■ File map

File	Contents
01-numbers-and-limits.md	Every number worth memorizing, in one place
02-mnemonics.md	All the list-recall tricks
03-compute.md	EC2, Auto Scaling, ELB, Lambda, containers, Beanstalk
04-storage.md	S3, EBS, EFS, FSx, Glacier, Storage Gateway, Snow family
05-databases.md	RDS, Aurora, DynamoDB, ElastiCache, Redshift
06-networking.md	VPC, subnets, SG vs NACL, Route 53, CloudFront, Direct Connect
07-security-identity.md	IAM, STS, KMS, Secrets Manager, WAF, Shield, GuardDuty
08-integration-analytics.md	SQS, SNS, EventBridge, Step Functions, Kinesis, Athena, EMR, Glue
09-monitoring-governance.md	CloudWatch, CloudTrail, Config, CloudFormation, Organizations
10-decision-triggers.md	"If the question says X → answer Y"
11-rapid-fire.md	150 one-line flashcards
99-review-of-original-notes.md	What the older slide decks got wrong, and what they are missing

01 — Numbers and Limits

Drill rule: cover the right column. Say the number out loud. These are asked directly, with no scenario wrapper, and they are free marks.

S3

Cue	Answer
Max object size	5 TB
Max size in a single PUT	5 GB
Multipart upload recommended above	100 MB
Multipart upload required above	5 GB
Multipart part size range	5 MB to 5 GB (last part may be smaller)
Max parts in a multipart upload	10,000
Max objects in a bucket	Unlimited
Buckets per account (default)	100 (soft limit, raise to 1,000)
Standard durability	11 nines (99.99999999%)
Standard availability SLA	99.99%
Read requests per second per prefix	5,500 GET/HEAD
Write requests per second per prefix	3,500 PUT/COPY/POST/DELETE
Minimum storage duration — Standard-IA / One Zone-IA	30 days
Minimum storage duration — Glacier Instant Retrieval	90 days
Minimum storage duration — Glacier Flexible Retrieval	90 days
Minimum storage duration — Glacier Deep Archive	180 days
Minimum billable object size for IA and Glacier classes	128 KB
Days before a Standard to Standard-IA lifecycle transition is allowed	30 days

S3 Glacier retrieval times

Tier	Time	Storage class
Instant Retrieval	milliseconds	Glacier Instant Retrieval
Expedited	1 to 5 minutes	Glacier Flexible Retrieval

Tier	Time	Storage class
Standard	3 to 5 hours	Glacier Flexible Retrieval
Bulk	5 to 12 hours	Glacier Flexible Retrieval
Standard	within 12 hours	Glacier Deep Archive
Bulk	within 48 hours	Glacier Deep Archive

Mnemonic for Flexible Retrieval: **Express, Standard, Bulk** = **minutes, hours, half a day**. Retrieved data stays available for download for **24 hours**.

EBS

Cue	Answer
gp3 — baseline included	3,000 IOPS and 125 MB/s
gp3 — maximum	16,000 IOPS and 1,000 MB/s
gp2 — IOPS per GB	3 IOPS per GB , capped at 16,000
gp2 — burst for small volumes	3,000 IOPS
io1 / io2 — max IOPS	64,000 (io2 Block Express: 256,000)
io2 — IOPS-to-size ratio	500:1 (io1 is 50:1)
st1 throughput HDD — max throughput	500 MB/s
sc1 cold HDD — max throughput	250 MB/s
Max volume size (gp2, gp3, io1, io2, st1, sc1)	16 TiB
Minimum size for st1 and sc1	125 GiB
Volume status check interval	every 5 minutes
Where snapshots are stored	S3 , incrementally, and they are region-scoped
Multi-Attach support	io1 and io2 only , up to 16 instances , same AZ

Volume type cheat: gp3 is the general default. io2 means "I need guaranteed IOPS." st1 means "big sequential reads, like a data warehouse or log processing." sc1 is the cheapest, coldest, rarely touched option.

Lambda

Cue	Answer
Max execution duration	900 seconds, which is 15 minutes
Memory range	128 MB to 10,240 MB , in 1 MB steps
Where you get a full vCPU	at about 1,769 MB of memory

Cue	Answer
<code>/tmp</code> ephemeral disk	512 MB default, up to 10,240 MB
Synchronous payload limit, request and response	6 MB
Asynchronous (event) payload limit	256 KB
Deployment package, zipped, direct upload	50 MB
Deployment package, unzipped	250 MB
Container image size	10 GB
Default concurrent executions per region	1,000 (soft limit)
Environment variable total size	4 KB
Layers per function	5

SQS

Cue	Answer
Max message size	256 KB (use the Extended Client Library plus S3 for up to 2 GB)
Default retention	4 days
Retention range	60 seconds to 14 days
Default visibility timeout	30 seconds
Visibility timeout range	0 seconds to 12 hours
Long polling max wait	20 seconds
Delay queue max	15 minutes
FIFO throughput without batching	300 messages per second
FIFO throughput with batching, 10 per call	3,000 messages per second
FIFO high-throughput mode	70,000 or more messages per second
Standard queue throughput	nearly unlimited
In-flight messages, standard	120,000
In-flight messages, FIFO	20,000

Delivery guarantees: Standard is at-least-once with best-effort ordering. FIFO is exactly-once with strict ordering. Design for duplicates on standard queues.

Kinesis

Cue	Answer
Max record size, Data Streams	1 MB
Per-shard write	1 MB/s or 1,000 records/s
Per-shard read, shared fan-out	2 MB/s and 5 read calls/s
Enhanced fan-out	2 MB/s per shard per consumer
Default retention	24 hours
Max retention	365 days
Firehose buffer interval	60 to 900 seconds
Firehose minimum latency	about 60 seconds — near-real-time, not real-time

DynamoDB

Cue	Answer
Max item size	400 KB
What 1 RCU buys	one strongly consistent read per second of up to 4 KB (or two eventually consistent)
What 1 WCU buys	one write per second of up to 1 KB
Partition key max size	2,048 bytes
Sort key max size	1,024 bytes
Local secondary indexes per table	5 , and they must be created with the table
Global secondary indexes per table	20 (soft limit)
Query and Scan result page size	1 MB
BatchGetItem limit	100 items or 16 MB
BatchWriteItem limit	25 items or 16 MB
Transaction limit	100 items or 4 MB
DynamoDB Streams retention	24 hours
TTL deletion window	within 48 hours of expiry

RDS and Aurora

Cue	Answer
Automated backup retention, default	7 days
Automated backup retention, range	0 to 35 days (0 turns it off)

Cue	Answer
Read replicas per RDS instance	5 (Aurora allows 15)
Max RDS storage, most engines	64 TiB
Max Aurora cluster volume	128 TiB
How many copies Aurora keeps	6 copies across 3 AZs
Aurora write tolerance	survives the loss of 2 copies
Aurora read tolerance	survives the loss of 3 copies
Multi-AZ replication type	synchronous
Read replica replication type	asynchronous
Multi-AZ failover time	typically 60 to 120 seconds
Aurora failover time	typically under 30 seconds
Aurora Global Database cross-region lag	typically under 1 second
Aurora Global Database secondary regions	up to 5

The single most-tested RDS distinction: Multi-AZ is for **availability** — synchronous, a standby you cannot read from, automatic failover. Read replicas are for **scaling reads** — asynchronous, readable, can live in another region, promoted manually.

VPC

Cue	Answer
VPC CIDR block size range	/16 (65,536 addresses) down to /28 (16 addresses)
Reserved IPs per subnet	5
Which five are reserved	network address, VPC router, DNS, reserved for future use, broadcast — the first four and the last one
Usable IPs in a /24 subnet	251 (256 minus 5)
VPCs per region, default	5
Subnets per VPC	200
Security groups per network interface	5 (raise to 16)
Rules per security group	60 inbound and 60 outbound
Security groups per VPC	2,500
NACLs per VPC	200
Rules per NACL	20 (raise to 40)
Route tables per VPC	200

Cue	Answer
Routes per route table	50
Internet gateways per VPC	1
Elastic IPs per region	5
NAT gateway bandwidth	scales up to 100 Gbps
NAT gateway simultaneous connections per destination	55,000

/32 means exactly one IP address. **/0** means the entire network. **0.0.0.0/0** as a route destination means "everything else."

Load balancers

Cue	Answer
ALB — OSI layer	Layer 7, HTTP and HTTPS
NLB — OSI layer	Layer 4, TCP, UDP, TLS
Gateway Load Balancer — OSI layer	Layer 3, IP
ALB idle timeout, default	60 seconds
NLB idle timeout	350 seconds , not configurable
NLB static addressing	one static IP per AZ , and it supports Elastic IPs
ALB static addressing	none — you must use the DNS name
Minimum subnets for an internet-facing load balancer	2, in 2 different AZs
Subnet size a load balancer needs	at least /27 with 8 free addresses
ALB rules per listener	100
Cross-zone load balancing by default	ALB: on. NLB: off. Classic: off.
Deregistration delay (connection draining), default	300 seconds , range 0 to 3,600

CloudWatch

Cue	Answer
Basic monitoring interval	5 minutes , free
Detailed monitoring interval	1 minute , paid
High-resolution custom metric interval	1 second
Metric retention, 1-second data	3 hours
Metric retention, 1-minute data	15 days
Metric retention, 5-minute data	63 days

Cue	Answer
Metric retention, 1-hour data	455 days, which is 15 months
Log retention range	1 day to 10 years, or never expire
Metrics EC2 reports with no agent	CPU, network, disk I/O, status checks
Metrics that need the CloudWatch agent	memory, disk space used, swap, per-process
Alarm states	OK, ALARM, INSUFFICIENT_DATA

The classic trap: memory utilization and disk space used are **not** default EC2 metrics. They need the unified CloudWatch agent. Any answer claiming "memory metrics with no agent" is wrong.

Auto Scaling

Cue	Answer
Default cooldown	300 seconds
Default health check grace period	300 seconds
Health check types	EC2 (default), ELB, custom
Default termination order	oldest launch template or configuration , then the instance closest to the next billing hour , then random
Max instances per group	a soft limit you can raise — it is not a hard 20
Launch templates versus launch configurations	templates are current and versioned; configurations are legacy

EC2 pricing and tenancy

Cue	Answer
Reserved Instance terms	1 or 3 years
Savings Plans terms	1 or 3 years
Maximum Reserved Instance or Savings Plan discount	up to about 72%
Spot discount	up to about 90%
Spot interruption warning	2 minutes
Spot block, fixed duration	retired — never the right answer
On-Demand Capacity Reservation	reserve capacity in one AZ with no term commitment
Dedicated Instance	hardware isolated per account
Dedicated Host	you get the whole physical server , needed for socket or core-based licences
Billing granularity	per second , with a 60-second minimum

Everything else worth memorizing

Cue	Answer
Snowcone capacity	8 TB usable HDD, 14 TB SSD
Snowball Edge Storage Optimized	80 TB usable
Snowmobile capacity	up to 100 PB
Rough rule for choosing Snow over the network	more than about 10 TB on a slow link
Direct Connect dedicated port speeds	1, 10, and 100 Gbps
Direct Connect hosted port speeds	50 Mbps to 10 Gbps
Direct Connect lead time	weeks to months — never the answer to an urgent need
Site-to-Site VPN tunnels per connection	2 , for redundancy
Site-to-Site VPN throughput per tunnel	about 1.25 Gbps
Route 53 multivalue answer records returned	up to 8 healthy records
Route 53 health check failure threshold	3 consecutive failed checks by default
CloudFront default TTL	24 hours
CloudFront max TTL	1 year
CloudFront price classes	All, 200, 100 — 100 is cheapest with the smallest footprint
Effect of a Service Control Policy on the management account	none — SCPs never restrict the management account
KMS automatic key rotation, AWS-managed keys	once a year
Data size you can encrypt directly with KMS	up to 4 KB — above that use envelope encryption
IAM roles per account	1,000
IAM users per account	5,000
Groups one user can belong to	10
Managed policies per role, user, or group	10 (raise to 20)

02 — Mnemonics

These are for lists you have to produce from a blank page. Learn the trigger word first, then unpack it.

Well-Architected Framework — six pillars

"Our Six Pillars Really Cost Something"

Letter	Pillar
O	Operational Excellence
S	Security
P	Performance Efficiency
R	Reliability
C	Cost Optimization
S	Sustainability

Trusted Advisor — five check categories

"CPFSS" — or the sentence "Cheap Performance Fixes Secure Systems"

Letter	Category
C	Cost Optimization
P	Performance
F	Fault Tolerance
S	Security
S	Service Limits

Route 53 routing policies — eight of them

"SWF-GLM-GM" — say it as "Simple Weighted Failover, Geo Latency Multivalue, Geoproximity, IP-based"

Policy	Pick it when
Simple	One resource serves the whole record
Weighted	Split traffic by percentage — A/B tests, gradual migration
Failover	Active-passive disaster recovery
Geolocation	Route by where the user is — compliance, language, content licensing
Geoproximity	Route by where your resources are, with a bias dial to shift traffic

Policy	Pick it when
Latency	Route to the region that answers fastest
Multivalued answer	Return up to 8 healthy records, health-checked, client picks
IP-based	Route by the user's CIDR block, for known ISP ranges

The pair people confuse: Geolocation is about the **user's** location. Geoproximity is about **your resources'** location and lets you shift load with a bias.

S3 storage classes — cheapest to most expensive access

"Stand In One Zone, Glacier Instantly Freezes Deeply"

Class	Use it when
S3 Standard	Hot, frequently accessed
S3 Intelligent-Tiering	Access pattern is unknown or changing — let AWS move it
S3 Standard-IA	Infrequent access, still need it fast, multi-AZ
S3 One Zone-IA	Infrequent access, recreatable data, one AZ only, 20% cheaper
Glacier Instant Retrieval	Archive you still need in milliseconds, accessed once a quarter
Glacier Flexible Retrieval	Archive, minutes to hours is fine
Glacier Deep Archive	Compliance retention, 12 to 48 hours is fine, cheapest of all

The five things a default VPC gives you

"I Really Should Note Down" — Internet gateway, Route table, Security group, Network ACL, DHCP option set

Item	What it does
I nternet gateway	Attached, so the VPC can reach the internet
R oute table	Main table with 0.0.0.0/0 pointing at the internet gateway
S ecurity group	Default group, allows all outbound and allows inbound from itself
N etwork ACL	Default NACL, allows all inbound and all outbound
D HCP option set	The account's default set, associated with the VPC

Plus: every subnet in a default VPC auto-assigns public IPv4 addresses.

S3 encryption options

"**3, C, K, K**" — SSE-S3, SSE-C, SSE-KMS, DSSE-KMS, plus client-side

Option	Who holds the key	Who does the encrypting
SSE-S3	AWS manages everything, AES-256	S3
SSE-KMS	You manage the customer master key in KMS; AWS manages the data key	S3
SSE-C	You supply the key with every request; AWS never stores it	S3
DSSE-KMS	Same as SSE-KMS but applied twice, for strict regimes	S3
Client-side	You	Your application, before upload

Since January 2023, **SSE-S3 is applied to all new objects by default**. The old note that "S3 is not encrypted by default" is out of date.

The four EC2 instance lifecycle outcomes

"**Stop, Hibernate, Reboot, Terminate**" — and what survives each

Action	EBS root volume	Instance store	RAM	Public IPv4
Reboot	Kept	Kept	Kept	Kept
Stop	Kept	Lost	Lost	Released, new one on start
Hibernate	Kept	Lost	Saved to the root EBS volume	Released
Terminate	Deleted if DeleteOnTermination is true	Lost	Lost	Released

Reboot is the only action where the instance store and the public IP both survive. That is the exam's favourite distinction here.

Instance store versus EBS

"**Store is temporary, Block is durable.**"

Instance store	EBS
Physically attached to the host	Network-attached
Highest possible IOPS	Good, provisioned IOPS
Data lost on stop or terminate	Data persists

Instance store	EBS
Cannot be detached or snapshotted	Detachable, snapshottable
Free, included in instance price	Billed separately
Buffers, caches, scratch, temporary	Boot volumes, databases, anything you keep

Security group versus NACL

"Groups guard instances. Lists guard subnets."

Security group	Network ACL
Instance level (really the network interface)	Subnet level
Stateful — return traffic is automatically allowed	Stateless — you must allow the return traffic explicitly
Allow rules only	Allow and deny rules
All rules evaluated together	Rules evaluated in number order , first match wins
Default: deny all inbound, allow all outbound	Default NACL: allow everything. A custom NACL: deny everything.
Does not filter traffic between instances in the same subnet when the same SG is used	Does not filter traffic between instances in the same subnet at all

Two memory hooks: **stateful** = "it remembers", and **a custom NACL starts closed while the default NACL starts open**. Both of those get tested.

Disaster recovery strategies — slowest and cheapest to fastest and dearest

"Backup, Pilot, Warm, Hot" — or "BPWH: Big Problems Won't Hurt"

Strategy	RTO / RPO	What is running
Backup and restore	Hours	Nothing — just backups in another region
Pilot light	Tens of minutes	Core data replicating, servers off
Warm standby	Minutes	A scaled-down but working copy
Multi-site active-active	Near zero	Full capacity in both regions, serving traffic

RTO = how long you may be down. RPO = how much data you may lose. "RTO = Time" and "RPO = Point in the past."

Cloud migration strategies — the 7 Rs

"Retire, Retain, Relocate, Rehost, Replatform, Refactor, Repurchase"

R	Meaning
Retire	Turn it off, nobody needs it

R	Meaning
Retain	Leave it where it is, for now
Relocate	Move the hypervisor, as with VMware Cloud on AWS
Rehost	Lift and shift, unchanged
Replatform	Lift and reshape — for example self-managed MySQL to RDS
Refactor	Rewrite it cloud-native
Repurchase	Drop it and buy SaaS instead

The IAM policy evaluation order

"Deny beats everything, then allow, then default deny."

1. An **explicit Deny** anywhere always wins.
2. Otherwise, an **explicit Allow** grants access.
3. Otherwise, **implicit deny** — nothing is permitted unless something allowed it.

Plus the guard rails that can only subtract: **SCPs, permission boundaries, and session policies never grant anything; they only limit what an Allow can reach.**

The four CloudFormation building blocks you must name

"PMRO" — Parameters, Mappings, Resources, Outputs. Only **Resources** is required.

Kinesis family — pick by verb

Service	Verb
Data Streams	Ingest and keep, with custom real-time processing
Data Firehose	Deliver to S3, Redshift, OpenSearch, Splunk — no code
Managed Service for Apache Flink	Query the stream with SQL or Flink
Video Streams	Ingest video

"Streams if I write the consumer. Firehose if AWS writes it for me."

03 — Compute

EC2 — Elastic Compute Cloud

■ Cue → Answer

Cue	Answer
The seven instance states	pending, running, stopping, stopped, shutting-down, terminated, rebooting
What EC2 charges you for while stopped	nothing for compute , but you still pay for EBS volumes and any unattached Elastic IPs
Where instance metadata lives	http://169.254.169.254/latest/meta-data/
Where user data lives	http://169.254.169.254/latest/user-data/
Which metadata service version you should prefer	IMDSv2 — it is session-based and defends against server-side request forgery
User data runs when	once, at first boot , as root
Instance cannot reach the internet — check what	public IP or Elastic IP present? route table has 0.0.0.0/0 to the internet gateway? security group outbound? NACL both ways?
What determines placement on physical hardware	the placement group
Status check types	system status check (AWS infrastructure) and instance status check (your OS/config)
A failing system status check means	AWS's problem — stop and start the instance to migrate it to new hardware
A failing instance status check means	your problem — the OS, the network config, a full disk

■ Placement groups

Type	Shape	Pick it for
Cluster	All instances packed onto one rack in one AZ	Lowest latency, highest throughput — HPC, tightly coupled jobs. Single point of failure.
Spread	Each instance on distinct hardware, up to 7 per AZ	Maximum isolation — a small number of critical instances
Partition	Groups of instances in separate partitions, each on its own rack	Large distributed systems — HDFS, Cassandra, Kafka

Mnemonic: **Cluster = Close. Spread = Separate. Partition = Pods.**

■ Default SSH user names

AMI	User
Amazon Linux / Amazon Linux 2 / AL2023	<code>ec2-user</code>
RHEL	<code>ec2-user</code> or <code>root</code>
Ubuntu	<code>ubuntu</code>
CentOS	<code>centos</code> or <code>ec2-user</code>
Debian	<code>admin</code>
SUSE	<code>ec2-user</code> or <code>root</code>
Bitnami	<code>bitnami</code>

■ Purchasing options — the cost-optimization questions live here

Option	Pick it when the question says
On-Demand	Short, unpredictable, cannot be interrupted, first time running this workload
Reserved Instance	Steady state, known for 1 or 3 years, up to ~72% off. Standard RI = cheapest but locked. Convertible RI = you can change family.
Savings Plans	Steady spend but you want flexibility across instance family, size, region, and even Fargate and Lambda
Spot Instances	Fault-tolerant, stateless, interruptible — batch, CI, rendering, big data. Up to ~90% off, 2-minute interruption notice
Dedicated Host	Socket or core-based software licence , or you need visibility of the physical server
Dedicated Instance	Hardware isolated from other accounts, but AWS still controls placement
Capacity Reservation	You must be certain capacity exists in a specific AZ, with no term commitment

The cost trap: if a question mentions "licence tied to physical cores or sockets," the answer is **Dedicated Host**, not Dedicated Instance. If it mentions "cannot tolerate interruption," Spot is wrong no matter how cheap.

■ Instance family letters

Letter	Family	Shape
T	Burstable	Low baseline with CPU credits — dev, small web
M	General purpose	Balanced CPU and memory
C	Compute optimized	High CPU — batch, modelling, gaming servers
R, X, z	Memory optimized	Big RAM — in-memory databases, caches
I, D, H	Storage optimized	High local disk IOPS or throughput — NoSQL, warehouses

Letter	Family	Shape
P, G, Inf, Trn	Accelerated	GPU and ML accelerators

Auto Scaling

Cue	Answer
The three things every group needs	minimum, desired, maximum capacity
What defines the instances	a launch template (versioned) — launch configurations are legacy
To roll out a new AMI	create a new launch template version , then refresh the instances
Scaling policy types	target tracking, step scaling, simple scaling, scheduled, predictive
Easiest and most-recommended policy	target tracking — "keep average CPU at 50%"
For a known traffic pattern, like every Monday at 9am	scheduled scaling
For a repeating daily or weekly pattern learned from history	predictive scaling
Default cooldown	300 seconds
Default health check grace period	300 seconds
To stop instances being killed before the app is up	raise the health check grace period
To stop the group flapping up and down	raise the cooldown and widen the CloudWatch alarm threshold
To take an instance out for troubleshooting without it being replaced	put it in standby
To run a script before an instance is terminated	a lifecycle hook
Health check that notices the application, not just the OS	ELB health check
What makes a group resilient to an AZ failure	span multiple AZs — the group rebalances automatically

■ What Auto Scaling can scale beyond EC2 (Application Auto Scaling)

EC2 instances, **EC2 Spot Fleets**, **ECS service desired count**, **DynamoDB table and index capacity**, **Aurora read replica count**, and more.

The classic scenario: "Instances are being terminated and replaced in a loop." Cause: the application takes longer to boot than the health check grace period. Fix: increase the grace period, or use an ELB health check pointed at a real readiness endpoint.

Elastic Load Balancing

Cue	Answer
ALB operates at	Layer 7 — it can read HTTP paths, hosts, headers, query strings
NLB operates at	Layer 4 — TCP, UDP, TLS

Cue	Answer
Gateway Load Balancer operates at	Layer 3 — it fronts third-party security appliances
Classic Load Balancer	legacy — only pick it if the question mentions EC2-Classic
Need a static IP or an Elastic IP on the load balancer	NLB
Need millions of requests per second and extreme low latency	NLB
Need to route /api to one target group and /images to another	ALB path-based routing
Need to route api.example.com separately from www.example.com	ALB host-based routing
Need to load balance containers with dynamic ports	ALB
Need WebSockets or HTTP/2	ALB
Need to authenticate users at the load balancer	ALB with Cognito or OIDC
Need to send a fixed response or redirect HTTP to HTTPS	ALB listener rule
Need UDP	NLB
Need to preserve the client's source IP	NLB does it natively; ALB passes it in X-Forwarded-For
Sticky sessions are implemented with	a cookie — AWSALB for ALB, duration-based or application-based
Health check sits where	on the target group
Where SSL/TLS termination happens	on the listener , using a certificate from ACM
To serve multiple certificates on one listener	SNI (Server Name Indication)
Cross-zone load balancing	on by default for ALB , off by default for NLB and Classic
Minimum subnets for an internet-facing load balancer	2 subnets in 2 different AZs

Lambda

Cue	Answer
Max duration	15 minutes — if a question needs longer, the answer is Fargate, ECS, Batch, or Step Functions
Memory	128 MB to 10,240 MB ; CPU scales with memory
How to make a function faster without changing code	give it more memory — you get proportionally more CPU
How to remove cold starts for a latency-sensitive function	provisioned concurrency
How to stop one function starving the account	reserved concurrency
Default regional concurrency	1,000

Cue	Answer
What a 429 TooManyRequestsException means	you hit a concurrency limit
Where failed asynchronous invocations go	a dead-letter queue (SQS or SNS) or an on-failure destination
To share code or dependencies between functions	Lambda layers
To run Lambda inside a VPC	attach it to subnets and a security group ; outbound internet then needs a NAT gateway
Lambda key metrics	Invocations, Errors, Duration, Throttles, DeadLetterErrors, ConcurrentExecutions
Lambda event sources, common ones	API Gateway, S3, DynamoDB Streams, Kinesis, SQS, SNS, EventBridge, CloudFront (Lambda@Edge), ALB
Lambda@Edge versus CloudFront Functions	Lambda@Edge for heavier logic with network access; CloudFront Functions for very light, very fast header and URL rewrites

Containers

Cue	Answer
ECS	AWS's own container orchestrator
EKS	Managed Kubernetes — pick it when the question says Kubernetes, kubectl, or "existing on-prem k8s"
Fargate	Serverless container compute — no servers to patch, size, or scale
EC2 launch type	You manage the container instances — pick it when the question needs GPU, specific instance types, or host access
ECR	The container registry. Images are region-scoped , so you must replicate or copy them to another region
ECS task role versus task execution role	Task role = permissions your container code uses. Task execution role = permissions ECS needs to pull the image and write logs.
To put a load balancer in front of ECS with dynamic ports	ALB
To reduce container management overhead	Fargate

Elastic Beanstalk

Cue	Answer
What it does	Handles capacity provisioning, load balancing, auto scaling, and health monitoring from your uploaded code
Platforms	Java, .NET, PHP, Node.js, Python, Ruby, Go, Docker
Environment types	Web Server environment and Worker environment
What a Worker environment is for	pulling from an SQS queue for background jobs

Cue	Answer
Deployment policies	All at once, Rolling, Rolling with additional batch, Immutable, Blue/Green (traffic splitting)
Zero-downtime, safest deployment	Immutable or Blue/Green
Cheapest and fastest but has downtime	All at once
Who owns the underlying resources	you do — Beanstalk creates them in your account and you can see and tune them

Beanstalk versus CloudFormation versus OpsWorks: Beanstalk = "deploy my app, you figure out the infrastructure."
CloudFormation = "build exactly the infrastructure I declared." OpsWorks = "I already use Chef or Puppet."

AWS Batch versus Lambda versus Fargate

Cue	Answer
Thousands of batch jobs, each possibly hours long	AWS Batch
Short event-driven function, under 15 minutes	Lambda
Long-running container, no servers to manage	Fargate
Needs to coordinate many steps with retries and branching	Step Functions

04 — Storage

The first question to ask on any storage question

Shape of the need	Service
A single instance needs a disk	EBS
Many instances need the same files, Linux	EFS
Many instances need the same files, Windows and SMB	FSx for Windows File Server
Objects over HTTP, any size, from anywhere	S3
Archive , retrieval in minutes to days	S3 Glacier classes
Temporary scratch at the highest possible IOPS	Instance store
High-performance computing shared scratch	FSx for Lustre
On-premises apps need cloud storage behind them	Storage Gateway
Moving petabytes with no bandwidth	Snow family

S3 — Simple Storage Service

■ Cue → Answer

Cue	Answer
Object size range	0 bytes to 5 TB
Single PUT limit	5 GB
Multipart upload range	parts of 5 MB to 5 GB , up to 10,000 parts
Namespace scope of bucket names	globally unique across all AWS accounts
Consistency model	strong read-after-write consistency for all operations, since December 2020
Default encryption	SSE-S3 (AES-256) is applied to all new objects automatically
Durability	11 nines
To protect against accidental deletes	versioning
To require MFA before a permanent delete	MFA Delete , which needs versioning and can only be enabled by the root user with the CLI
To make objects undeletable for a fixed period	Object Lock in governance or compliance mode, plus a legal hold. Needs versioning.
To copy objects automatically to another region	Cross-Region Replication (CRR)
To copy objects automatically within a region	Same-Region Replication (SRR)

Cue	Answer
What replication requires	versioning on both buckets , plus an IAM role granting S3 permission to replicate
Does replication copy existing objects	no, only new ones — use S3 Batch Replication for what is already there
To move data between classes on a schedule	lifecycle rules
To let a browser upload straight to S3	a presigned URL or presigned POST
To serve a static website	enable static website hosting ; for HTTPS and a custom domain put CloudFront in front
To stop all public access account-wide	S3 Block Public Access
The modern way to let CloudFront read a private bucket	Origin Access Control (OAC) — Origin Access Identity (OAI) is the legacy name
To run SQL against a single object	S3 Select
To run SQL across many objects and other sources	Athena
Events S3 can publish	to SQS, SNS, Lambda, and EventBridge
To speed up uploads from far away	S3 Transfer Acceleration — uses CloudFront edge locations
To get a report of what is in a bucket	S3 Inventory
To find out which objects are costing you and how they are accessed	S3 Storage Lens and Storage Class Analysis
To do a bulk operation over millions of objects	S3 Batch Operations
To find personal data in a bucket	Amazon Macie

■ Storage classes

Class	Availability	AZs	Minimum duration	Retrieval fee	Use
Standard	99.99%	3+	none	none	Hot data
Intelligent-Tiering	99.9%	3+	none	none (small monitoring fee)	Unknown or shifting patterns
Standard-IA	99.9%	3+	30 days	yes	Infrequent, needs multi-AZ
One Zone-IA	99.5%	1	30 days	yes	Infrequent, re-creatable
Glacier Instant Retrieval	99.9%	3+	90 days	yes	Archive, millisecond access
Glacier Flexible Retrieval	99.99%	3+	90 days	yes	Archive, minutes to hours

Class	Availability	AZs	Minimum duration	Retrieval fee	Use
Glacier Deep Archive	99.99%	3+	180 days	yes	Compliance archive, 12 to 48 hours

The One Zone-IA trap: it is cheaper because it lives in **one AZ**. If the question says "must survive the loss of an Availability Zone," One Zone-IA is wrong. It is only right for data you could regenerate, like thumbnails or transcodes.

■ Lifecycle rule constraints worth memorizing

- Standard to Standard-IA or One Zone-IA: **minimum 30 days** in Standard first.
- You can transition **down** the chain, never **up** — there is no lifecycle transition from Glacier back to Standard. To get it back you **restore** a copy.
- Objects smaller than **128 KB** are not transitioned to IA or Glacier by lifecycle rules.
- You can expire **current** versions, expire **noncurrent** versions, and **clean up incomplete multipart uploads** — that last one is a standard cost-saving answer.

■ Who wins on S3 access — policy evaluation

1. An **explicit Deny** in any policy wins — IAM policy, bucket policy, SCP, VPC endpoint policy.
2. **Block Public Access** overrides any bucket policy or ACL that grants public access.
3. Otherwise, an **Allow** in **either** the IAM policy **or** the bucket policy is enough, as long as both accounts are the same. Cross-account needs an Allow on **both** sides.
4. ACLs are **disabled by default** on new buckets and AWS recommends leaving them off.

EBS — Elastic Block Store

Cue	Answer
Scope	one Availability Zone — a volume cannot cross AZs
To move a volume to another AZ	snapshot it, then create a volume from the snapshot in the target AZ
To move to another region	snapshot, copy the snapshot to the region, create the volume
Attachment	normally one instance at a time ; io1 and io2 support Multi-Attach up to 16 instances in one AZ
Replication	automatically replicated within its AZ
What you can change on a live volume	type, size, and provisioned IOPS — elastic volumes, no downtime
Can you shrink a volume	no — only grow
Snapshots are	incremental , stored in S3 , and region-scoped
CLI command to snapshot	<code>aws ec2 create-snapshot</code>
To automate snapshot schedules and retention	Data Lifecycle Manager (DLM) or AWS Backup
Encryption	AES-256 via KMS ; snapshots of an encrypted volume are encrypted; you can encrypt on restore

Cue	Answer
Can you encrypt an existing unencrypted volume in place	no — snapshot it, copy the snapshot with encryption on, create a new volume
Volume status values	ok, impaired, insufficient-data , checked every 5 minutes

■ Volume types — pick by the phrase in the question

Phrase in the question	Volume type
"general purpose", "cost-effective", default	gp3 (prefer it over gp2 — cheaper and faster baseline)
"sub-millisecond", "guaranteed IOPS", "critical database", "> 16,000 IOPS"	io2 or io2 Block Express
"large sequential reads", "big data", "log processing", "throughput"	st1
"lowest cost", "infrequently accessed", "cold"	sc1
"boot volume"	gp3, gp2, io1, io2 — st1 and sc1 cannot be boot volumes
"highest IOPS possible, data is temporary"	instance store , not EBS

EFS — Elastic File System

Cue	Answer
Protocol	NFS v4.1
Operating systems	Linux only
Scope	regional — mount targets in multiple AZs, so it survives an AZ loss
Concurrency	thousands of instances at once , shared read-write
Capacity	grows and shrinks automatically , pay for what you use
Storage classes	Standard, One Zone, Standard-IA, One Zone-IA
To move cold files to IA automatically	lifecycle management
Performance modes	General Purpose (default, lowest latency) and Max I/O (higher throughput, higher latency)
Throughput modes	Bursting, Elastic, Provisioned
Encryption	at rest with KMS , in transit with TLS
Access control	IAM , plus security groups on the mount targets , plus EFS Access Points

EFS versus EBS in one line: EBS is one disk for one instance in one AZ. EFS is one file system for many instances across many AZs.

FSx

Flavour	Pick it when
FSx for Windows File Server	SMB , Windows, Active Directory integration, NTFS permissions, DFS
FSx for Lustre	HPC , machine learning, hundreds of GB/s, and it can link to an S3 bucket
FSx for NetApp ONTAP	You need ONTAP features — snapshots, cloning, multi-protocol NFS and SMB
FSx for OpenZFS	Moving ZFS workloads, NFS, snapshots

If the question says **Windows** or **SMB** or **Active Directory**, it is **FSx for Windows**, never EFS.

Storage Gateway — hybrid

Gateway type	What on-premises sees	Where data lands
S3 File Gateway	NFS or SMB file share	S3 objects
FSx File Gateway	SMB share with local cache	FSx for Windows
Volume Gateway — cached	iSCSI disk	Primary data in S3 , frequently used data cached locally
Volume Gateway — stored	iSCSI disk	Primary data stays local , backed up asynchronously to S3 as EBS snapshots
Tape Gateway	Virtual tape library (VTL)	S3 and Glacier , replacing physical tape

Cached versus stored, in one question each. "We want to shrink our on-premises storage footprint" → **cached**. "We need the whole dataset on site with low latency, backed up to AWS" → **stored**. "We want to retire our tape backup robot" → **Tape Gateway**.

Snow family and DataSync

Cue	Answer
Snowcone	Smallest, 8 TB HDD or 14 TB SSD, rugged, can run edge compute
Snowball Edge Storage Optimized	80 TB usable — the standard bulk transfer answer
Snowball Edge Compute Optimized	Edge processing with GPU options
Snowmobile	Up to 100 PB in a shipping container — exabyte-scale datacentre evacuation
AWS DataSync	Online transfer and ongoing sync between on-premises NFS/SMB/HDFS/object storage and S3, EFS, or FSx. Needs bandwidth.
AWS Transfer Family	Managed SFTP, FTPS, FTP front door onto S3 or EFS

Cue	Answer
Rule of thumb	If the question stresses limited or no bandwidth , or a hard deadline with petabytes , pick Snow . If it stresses ongoing, scheduled, incremental sync , pick DataSync .

AWS Backup

Cue	Answer
What it is	One place to define backup plans, schedules, retention, and lifecycle across services
Services it covers	EBS, EC2, RDS, Aurora, DynamoDB, EFS, FSx, Storage Gateway, S3, DocumentDB, Neptune
Features that answer compliance questions	cross-region copy, cross-account copy, Backup Vault Lock (write-once-read-many), audit reports
When to pick it over DLM	when the question says multiple services, central policy, or compliance reporting

AMIs — Amazon Machine Images

Cue	Answer
Scope	region-scoped — to use one in another region you must copy it there
To share with another account	modify the AMI's launch permissions ; encrypted snapshots also need the KMS key shared
To standardize and schedule image builds	EC2 Image Builder
Backing types	EBS-backed (can be stopped, persists) and instance-store-backed (cannot be stopped)

05 — Databases

Pick the database first

Shape of the need	Service
Relational, standard engine, managed	RDS
Relational, needs MySQL or PostgreSQL compatibility with cloud-scale performance	Aurora
Relational, want to pay nothing when idle	Aurora Serverless v2
Key-value or document, single-digit millisecond, massive scale	DynamoDB
DynamoDB but needs microseconds	DynamoDB + DAX
Caching in front of a database	ElastiCache (Redis or Memcached)
Petabyte analytics and reporting over structured data	Redshift
SQL over data already sitting in S3, no servers	Athena
Graph — social, fraud rings, recommendations	Neptune
Time series — IoT sensors, metrics	Timestream
Ledger with a cryptographically verifiable history	QLDB
MongoDB-compatible document store	DocumentDB
Wide-column Cassandra-compatible	Keyspaces
In-memory durable data store, Redis-compatible	MemoryDB for Redis

RDS — Relational Database Service

Cue	Answer
Engines	Aurora, MySQL, PostgreSQL, MariaDB, Oracle, Microsoft SQL Server
What AWS manages	provisioning, patching, backups, recovery, failure detection, repair
What you do not get	operating system access — no SSH to the host
If a question requires OS access or an unsupported engine	run the database on EC2 , not RDS
Automated backup retention, default and range	7 days , range 0 to 35 days
What automated backups let you do	point-in-time recovery , usually to within 5 minutes
Manual snapshots	kept until you delete them , and they survive deleting the instance
Restoring a snapshot produces	a new instance with a new endpoint — it never restores in place

Cue	Answer
Can you encrypt an existing unencrypted RDS instance	no, not in place — snapshot it, copy the snapshot with encryption on, restore from that
Encryption at rest uses	KMS
Encryption in transit uses	SSL/TLS
Maintenance window	a weekly window when AWS applies patches; Multi-AZ patches the standby first, then fails over

■ Multi-AZ versus read replicas — the most-tested table on the exam

	Multi-AZ	Read replica
Purpose	High availability	Read scaling
Replication	Synchronous	Asynchronous
Can you read from it	No	Yes
Failover	Automatic , 60 to 120 seconds	Manual promotion
Location	Another AZ in the same region	Same AZ, another AZ, or another region
How many	1 standby (or 2 readable standbys in the newer Multi-AZ DB cluster)	5 per instance, 15 for Aurora
Endpoint	Same endpoint , DNS flips	Its own endpoint
What it protects against	AZ failure, instance failure, patching downtime	nothing — it is not a HA feature

Say it out loud: "Multi-AZ is durability and availability. Read replicas are scalability." Both can be true at once — you can have Multi-AZ **and** read replicas.

■ Multi-AZ failover triggers

AZ outage, primary instance failure, instance type change, OS patching, manual reboot with failover, storage failure. **Not** triggered by heavy read load.

Aurora

Cue	Answer
Compatibility	MySQL and PostgreSQL
Performance claim	up to 5x MySQL, 3x PostgreSQL
Storage	grows automatically in 10 GB steps up to 128 TiB
Copies of your data	6 copies across 3 AZs
Fault tolerance	survives losing 2 copies for writes, 3 copies for reads , self-healing

Cue	Answer
Read replicas	up to 15 Aurora Replicas , with sub-10-millisecond replica lag
Failover time	typically under 30 seconds
Endpoints	cluster (writer) , reader , and custom endpoints
To scale replicas automatically	Aurora Auto Scaling on the replica count
For cross-region, near-zero-lag disaster recovery	Aurora Global Database — up to 5 secondary regions, typically under 1 second lag, failover in about 1 minute
For unpredictable or intermittent load	Aurora Serverless v2 — scales in fine increments, can scale to a low floor
To run analytics without touching the main cluster	Aurora parallel query or a reader endpoint
Zero-ETL to Redshift	Aurora zero-ETL integration with Redshift
Backtrack	rewind an Aurora MySQL cluster in place to a recent point, without restoring a snapshot
Cloning	fast database cloning — a copy-on-write clone, far quicker and cheaper than a snapshot restore

DynamoDB

Cue	Answer
Type	serverless key-value and document NoSQL
Latency	single-digit milliseconds at any scale
Max item size	400 KB
Primary key options	partition key alone , or partition key + sort key (composite)
Capacity modes	on-demand (unpredictable traffic, pay per request) and provisioned (predictable traffic, cheaper, supports auto scaling)
Read consistency	eventually consistent by default ; ask for strongly consistent and it costs twice the RCUs
1 RCU	one strongly consistent read per second of up to 4 KB
1 WCU	one write per second of up to 1 KB
Microsecond reads	DAX — a fully managed, highly available, in-memory cache that sits in front of the table
Is DAX good for write-heavy work	no — it is a read-through cache
Global secondary index	different partition and sort key , can be added any time, has its own capacity, eventually consistent only
Local secondary index	same partition key, different sort key , must be created with the table , shares the table's capacity, can be strongly consistent
To replicate a table across regions, active-active	Global Tables

Cue	Answer
To react to every change to an item	DynamoDB Streams plus Lambda, 24-hour retention
To delete old items automatically and for free	TTL on an attribute holding an epoch timestamp
To stop two writers clobbering each other	conditional writes / optimistic locking with a version attribute
To make several writes all-or-nothing	transactions
To back it up	point-in-time recovery (35 days) and on-demand backups
Scan versus Query	Query uses the key and is efficient. Scan reads the whole table and is the thing to avoid.
Symptom: ProvisionedThroughputExceededException	a hot partition or under-provisioned capacity — fix the key design or switch to on-demand
How to design a good partition key	high cardinality, evenly accessed — user ID, not a status flag

ElastiCache

	Redis	Memcached
Data structures	Lists, sets, sorted sets, hashes, bitmaps	Simple key-value only
Persistence	Yes , snapshots and AOF	No
Replication and failover	Yes , Multi-AZ with automatic failover	No
Read replicas	Yes	No
Multi-threaded	No (mostly single-threaded)	Yes
Scaling shape	Sharding (cluster mode) and replicas	Horizontal, add nodes
Pub/sub, geospatial, Lua	Yes	No
Pick it when the question says	"high availability", "persistence", "leaderboard", "session store that must survive", "sorted set"	"simplest possible cache", "multi-threaded", "no persistence needed", "scale out horizontally"

Caching concept	Meaning
Lazy loading / cache-aside	Only cache on a miss. Cheap on memory, but the first request is always slow and data can go stale.
Write-through	Write to cache and database together. Cache is always fresh, but you cache data nobody reads.
TTL	Expire entries, the usual answer to "how do we stop stale data?"

The classic scenario: "A read-heavy relational workload is slowing down and the same queries repeat." The two valid answers are **a read replica** (scale reads at the database) and **ElastiCache** (remove the reads entirely). If the question says "reduce database load" and "microsecond or sub-millisecond", pick the cache.

Redshift

Cue	Answer
Type	Petabyte-scale columnar data warehouse , for OLAP not OLTP
Architecture	one leader node that parses and distributes SQL, plus compute nodes with slices
What the leader node does	receives queries, builds the plan, distributes SQL to the compute nodes when the query touches user tables
Availability shape	historically single-AZ ; now supports Multi-AZ for RA3 clusters
Backups	automatic snapshots to S3, and cross-region snapshot copy for disaster recovery
To query data left in S3 without loading it	Redshift Spectrum
To control concurrency and give short queries priority	Workload Management (WLM) — it defines query queues
To run without managing clusters	Redshift Serverless
Distribution styles	KEY (co-locate join keys), EVEN , ALL (copy a small dimension table to every node), AUTO
Sort keys	compound (default, best for predictable filter order) and interleaved (equal weight across columns)
To load data efficiently	the COPY command from S3, in parallel
Encryption	KMS or CloudHSM, and enhanced VPC routing keeps traffic inside your VPC
Redshift versus Athena	Redshift for repeated, complex, high-concurrency reporting on loaded data. Athena for ad-hoc SQL on S3 with nothing to run.
Redshift versus EMR	Redshift for SQL analytics. EMR for Hadoop, Spark, and custom processing frameworks.

The purpose-built databases

Service	One-line trigger
Neptune	"graph", "relationships", "social network", "fraud ring", "knowledge graph", "Gremlin", "SPARQL"
Timestream	"time series", "IoT sensor readings", "metrics over time"
QLDB	"immutable", "cryptographically verifiable", "complete history of changes", "central trusted ledger"
Managed Blockchain	"multiple parties", "no central trusted authority", "Hyperledger", "Ethereum"
DocumentDB	"MongoDB compatible", "JSON documents", managed
Keyspaces	"Cassandra compatible", "CQL"
MemoryDB for Redis	"in-memory speed and durability as the primary database"

QLDB versus Managed Blockchain: QLDB has **one owner** you trust. Managed Blockchain has **several parties** who do not fully trust each other.

Database migration

Cue	Answer
Move a database to AWS with minimal downtime	AWS DMS — the source stays live during migration
Change engines, for example Oracle to Aurora PostgreSQL	DMS plus the Schema Conversion Tool (SCT)
Keep source and target in sync for a while	DMS ongoing replication (change data capture)
Move whole servers, not just databases	AWS Application Migration Service (MGN)

06 — Networking and Content Delivery

VPC fundamentals

Cue	Answer
Scope of a VPC	one region , spanning all its Availability Zones
Scope of a subnet	one Availability Zone — a subnet never spans AZs
CIDR block size range	/16 down to /28
Reserved addresses per subnet	5 — the first four and the last
Can you change a VPC's primary CIDR	no , but you can add secondary CIDR blocks
What makes a subnet public	a route to an internet gateway in its route table. Nothing else.
What an instance in a public subnet also needs	a public IP or Elastic IP
Internet gateways per VPC	1
What lets private instances reach out to the internet	a NAT gateway (or legacy NAT instance) in a public subnet
Can anything on the internet start a connection through a NAT gateway	no — outbound only
NAT gateway availability	AZ-scoped — deploy one per AZ for resilience, and route each private subnet to the one in its own AZ
NAT gateway versus NAT instance	Gateway is managed, scales to 100 Gbps, highly available in its AZ. Instance is an EC2 you patch, and you must disable the source/destination check .
Egress-only internet gateway	the IPv6 equivalent of a NAT gateway
Default VPC subnets	one public subnet per AZ, auto-assigning public IPv4

■ The five things a default VPC comes with

1. An **internet gateway** attached.
2. A **main route table** with `0.0.0.0/0` pointing at that internet gateway.
3. A **default security group**, allowing all outbound and allowing inbound from itself.
4. A **default network ACL**, allowing all inbound and all outbound.
5. The account's **default DHCP option set**, associated.

Security groups versus NACLs

	Security group	Network ACL
Level	Instance / network interface	Subnet
State	Stateful — return traffic allowed automatically	Stateless — you must allow return traffic yourself
Rules	Allow only	Allow and Deny

	Security group	Network ACL
Evaluation	All rules together	In rule-number order, lowest first, first match wins
Default behaviour	Deny all inbound, allow all outbound	Default NACL allows everything; a custom NACL denies everything
Can reference another security group	Yes	No — IP ranges only
Changes take effect	Immediately	Immediately
Filters traffic between instances in the same subnet	Yes, if they are in different security groups	No
How many apply	up to 5 per interface , all combined	exactly 1 per subnet

The ephemeral port trap. Because NACLs are stateless, an inbound rule for port 443 is not enough — you also need an **outbound** rule for the ephemeral port range (**1024 to 65535**) so the response can get out. SSH through a NACL needs **both** an inbound and an outbound rule.

The blocking trap. To block a single malicious IP address, you must use a **NACL deny rule**. A security group cannot deny anything.

One-line summary for the exam: security groups usually control **which ports** are open on your instances; NACLs usually control **which networks or IP ranges** can reach a subnet.

Connecting VPCs and networks

Cue	Answer
Connect two VPCs privately	VPC peering
Is peering transitive	No — if A peers B and B peers C, A cannot reach C
Can peered VPCs have overlapping CIDRs	No
Does peering cross regions and accounts	Yes, both
Connect hundreds of VPCs and on-premises networks	Transit Gateway — a hub-and-spoke router, and it is transitive
Share one VPC's subnets with other accounts	VPC sharing via AWS Resource Access Manager (RAM)
Expose your own service to other VPCs without peering	AWS PrivateLink / VPC endpoint services with an NLB
Encrypted tunnel from your datacentre over the internet	Site-to-Site VPN — 2 tunnels, about 1.25 Gbps each, set up in hours
Private, dedicated, consistent-bandwidth link to AWS	Direct Connect — 1, 10, or 100 Gbps, takes weeks to months
Highest-resilience hybrid design	Direct Connect with a Site-to-Site VPN as backup , or two Direct Connect links at different locations
Encrypt a Direct Connect link	run a VPN over the Direct Connect public virtual interface , or use MACsec
Connect individual users or laptops	AWS Client VPN
Component on the AWS side of a VPN	a virtual private gateway attached to the VPC

Cue	Answer
Component on your side of a VPN	a customer gateway

The hybrid question's decision line: if the question says "**urgent**", "**within days**", "**temporary**" → **Site-to-Site VPN**. If it says "**consistent bandwidth**", "**low jitter**", "**large steady transfers**", "**private, not over the internet**" → **Direct Connect**. If it says **both resilience and private bandwidth** → **Direct Connect with VPN failover**.

VPC endpoints

Type	Works with	How it works
Gateway endpoint	S3 and DynamoDB only	A route in your route table . Free.
Interface endpoint (PrivateLink)	Almost every other AWS service, plus partner and your own services	An elastic network interface with a private IP in your subnet, protected by a security group . Charged per hour and per GB.

Cue	Answer
Why use an endpoint at all	traffic to the AWS service never leaves the AWS network — no internet gateway, NAT, VPN, or Direct Connect needed
Are endpoints highly available	yes — horizontally scaled, redundant, no bandwidth bottleneck
To restrict which buckets can be reached through an endpoint	a VPC endpoint policy
Private subnet needs S3 but has no NAT gateway	gateway endpoint for S3 — and this is also the cost-saving answer, because it removes NAT data processing charges

Route 53

Cue	Answer
What it is	a highly available authoritative DNS service, plus domain registration and health checks
Hosted zone types	public (internet-resolvable) and private (resolvable only inside associated VPCs)
Alias record versus CNAME	Alias works at the zone apex (example.com), is free , and points at AWS resources. CNAME cannot be used at the apex and is charged per query.
What an Alias can point at	ELB, CloudFront, API Gateway, S3 static website, Global Accelerator, another Route 53 record, Elastic Beanstalk, VPC endpoint
What an Alias cannot point at	an EC2 instance DNS name — use an A record to its IP
TTL on an Alias to an ELB	you do not set it — Route 53 uses the target's
Record types you should recognize	A, AAAA, CNAME, MX, TXT, NS, SOA, SRV, PTR, CAA, Alias
Health checks can watch	an endpoint, other health checks (calculated) , or a CloudWatch alarm

Cue	Answer
Default health check failure threshold	3 consecutive failures
Resolve on-premises DNS names from a VPC and vice versa	Route 53 Resolver inbound and outbound endpoints
To block DNS queries to malicious domains	Route 53 Resolver DNS Firewall

■ Routing policies — the trigger table

The question says	Policy
One resource, nothing clever	Simple
"Send 10% to the new version", "A/B test", "gradual migration"	Weighted
"Active-passive", "disaster recovery site", "fail over to a static S3 page"	Failover
"Users in Germany must see the German site", "content licensing by country", "block a country"	Geolocation
"Shift traffic from one region to another with a bias"	Geoproximity (needs traffic flow)
"Serve users from the fastest region"	Latency
"Return several healthy IPs and let the client choose", "basic load balancing in DNS"	Multivalue answer
"Route based on the user's ISP CIDR block"	IP-based

Geolocation versus Latency: geolocation is about **compliance and content**, latency is about **speed**. If the question says "lowest latency," never pick geolocation.

CloudFront

Cue	Answer
What it is	a content delivery network of edge locations that caches and accelerates content
What it accelerates besides static files	dynamic content, APIs, and whole websites , via persistent connections over the AWS backbone
Origins it supports	S3, ALB, EC2, API Gateway, MediaStore, or any custom HTTP origin
Default TTL and max TTL	24 hours and 1 year
To control caching per request	cache policies and origin request policies (the legacy name is "cache behaviours with forwarded values")
To remove an object from the cache now	an invalidation — but versioned file names are the cheaper and recommended approach

Cue	Answer
To keep a private S3 bucket private while CloudFront serves it	Origin Access Control (OAC)
To restrict content to paying users	signed URLs (one file) and signed cookies (many files)
To block or allow whole countries	geo-restriction
To run code at the edge	CloudFront Functions (lightweight, sub-millisecond, header and URL work) and Lambda@Edge (heavier, can call other services)
To reduce cost by shrinking the footprint	a lower price class — All, 200, or 100
To add a WAF	attach a web ACL to the distribution
To use your own domain with HTTPS	an ACM certificate in us-east-1 — this region requirement is tested
The four viewer-facing layers that need us-east-1 ACM	CloudFront, and that is the one to remember

■ CloudFront versus Global Accelerator

	CloudFront	Global Accelerator
Caches content	Yes	No
Protocols	HTTP and HTTPS	TCP and UDP , any protocol
Gives you	a distribution domain name	two static anycast IP addresses
Pick it for	static and dynamic web content	non-HTTP traffic, gaming, VoIP, IoT, or when you need static IPs and fast regional failover

Elastic IPs and ENIs

Cue	Answer
Elastic IP	a static public IPv4 you own in a region, remappable between instances
When you are charged for an Elastic IP	when it is not associated with a running instance — and, since 2024, all public IPv4 addresses carry an hourly charge
Default Elastic IPs per region	5
Elastic network interface (ENI)	a virtual network card; it keeps its private IP, MAC address, and security groups when you move it
Use for a secondary ENI	a fixed management IP , a low-budget high-availability failover , or separating networks
ENI scope	one Availability Zone

VPC Flow Logs

Cue	Answer
What they capture	metadata about IP traffic to and from network interfaces — source, destination, ports, protocol, bytes, and ACCEPT or REJECT
What they do not capture	the packet contents — for that you need VPC Traffic Mirroring
Levels you can enable them at	VPC, subnet, or network interface
Where they can go	CloudWatch Logs, S3, or Kinesis Data Firehose
Classic use	diagnosing why traffic is blocked — a REJECT tells you a security group or NACL is in the way
Traffic flow logs never capture	Amazon DNS server queries, Windows licence activation, instance metadata at 169.254.169.254, DHCP traffic

Bastion hosts and Systems Manager

Cue	Answer
Traditional way to reach a private instance	a bastion host in a public subnet, with a security group allowing SSH only from your office CIDR
Modern way with no bastion, no open ports, no keys	AWS Systems Manager Session Manager
Why Session Manager is usually the better answer	no inbound ports, no SSH keys to manage, all sessions logged to CloudTrail and CloudWatch
What a private instance needs for Session Manager	the SSM Agent and an instance profile with the SSM policy, plus either a NAT gateway or VPC interface endpoints for SSM

07 — Security and Identity

This is the **30% domain** — the largest on the exam. When two answers both work, the one with less standing privilege usually wins.

IAM

Cue	Answer
User	A person or a service with long-term credentials
Group	A collection of users. Groups cannot contain other groups, and a group is not a principal — you cannot grant a group a role.
Role	An identity with no long-term credentials , assumed temporarily for a set of permissions
Policy	A JSON document stating what is allowed or denied
Scope of IAM	global — not regional
Best practice for the root user	enable MFA, create an admin IAM user or an Identity Center user, then never use root again. Delete root access keys.
Things only the root user can do	close the account, change the support plan, change the account name or email, restore IAM user permissions, enable MFA Delete on S3, register as a seller in the Reserved Instance Marketplace
How an EC2 instance should get credentials	an instance profile with an IAM role — never hard-coded access keys
How a Lambda function gets credentials	its execution role
How an application on-premises gets temporary credentials	IAM Roles Anywhere or STS with an identity provider
How to give a mobile app access	Cognito identity pools , which exchange a social or user-pool login for temporary credentials
How to let another AWS account's users in	a role with a trust policy naming that account , which they assume with sts:AssumeRole
How to centrally manage workforce sign-in across many accounts	IAM Identity Center (the successor to AWS SSO)
Policy evaluation order	explicit Deny wins → explicit Allow grants → otherwise implicit deny
What never grants permission, only limits it	SCPs, permission boundaries, session policies
To find unused permissions and over-broad policies	IAM Access Analyzer and the last-accessed data in the credential report
To audit all credentials in the account	the credential report

■ The policy elements you must recognize

Element	Meaning
Version	Always "2012-10-17"
Effect	Allow or Deny
Action	s3:GetObject , ec2:*
Resource	The ARN the action applies to
Principal	Only in resource-based policies — who is being granted access
Condition	aws:SourceIp , aws:MultiFactorAuthPresent , aws:PrincipalOrgID , s3:prefix , aws:RequestedRegion

Identity-based versus resource-based. Identity-based policies attach to a user, group, or role and have **no Principal**. Resource-based policies attach to the resource — an **S3 bucket policy, SQS queue policy, SNS topic policy, KMS key policy, Lambda resource policy** — and **must** have a Principal. Cross-account access usually needs both sides to agree.

■ ARN shape — recognize it on sight

```
arn:aws:service:region:account-id:resource
arn:aws:s3::my-bucket/my-key      <- S3 has no region or account
arn:aws:iam::123456789012:role/MyRole <- IAM is global, so no region
```

STS — Security Token Service

Cue	Answer
What it issues	temporary credentials — an access key, a secret key, and a session token
Main API calls	AssumeRole, AssumeRoleWithWebIdentity, AssumeRoleWithSAML, GetSessionToken, GetFederationToken
Session duration	15 minutes to 12 hours (1 hour default for most)
Use it for	cross-account access, federation, temporary elevated access, identity federation for mobile and web apps

KMS, CloudHSM, and encryption

Cue	Answer
KMS	Managed key service, integrated with nearly every AWS service. Multi-tenant, FIPS 140-2 Level 3 validated HSMs.
CloudHSM	A single-tenant, dedicated hardware security module you control. You manage the keys and AWS cannot see them.
When CloudHSM is the required answer	"FIPS 140-2 Level 3 single-tenant" , "we must be the only ones with key access", "we need to run our own PKCS#11 / JCE / CNG", "SQL Server or Oracle TDE with our own HSM"

Cue	Answer
KMS key types	AWS managed (free, rotated yearly, you cannot change the policy), customer managed (you control the policy and rotation), AWS owned
Max data you can encrypt directly with KMS	4 KB
For anything bigger	envelope encryption — KMS gives you a data key, you encrypt the data with it, and store the encrypted data key alongside
API call that does that	GenerateDataKey
Key rotation	automatic yearly for AWS managed keys; optional, configurable for customer managed keys. The old key material is kept so old ciphertext still decrypts.
Scope of a KMS key	regional — to use encrypted snapshots in another region you must re-encrypt with a key in that region (or use a multi-Region key)
Who controls access to a KMS key	the key policy , plus IAM policies and grants. A key policy is required — IAM alone is not enough.
To delete a key	schedule deletion with a 7 to 30 day waiting period
To store passwords and database credentials with automatic rotation	Secrets Manager
To store configuration and plain parameters for free	Systems Manager Parameter Store (SecureString for encrypted values)
Secrets Manager versus Parameter Store	Secrets Manager costs money but gives built-in rotation with Lambda and cross-account access . Parameter Store is free up to the standard tier but has no automatic rotation .
To issue and renew TLS certificates for free	ACM — public certificates auto-renew
Where an ACM certificate must live for CloudFront	us-east-1

The protection services

Service	What it does	Trigger phrase
AWS WAF	Layer 7 web firewall on CloudFront, ALB, API Gateway, AppSync, Cognito	"SQL injection" , "cross-site scripting" , "block by rate per IP", "block by country at layer 7", "block a specific URI pattern"
AWS Shield Standard	Free, automatic layer 3 and 4 DDoS protection for everyone	"DDoS", no cost mentioned
AWS Shield Advanced	Paid, 24/7 DDoS response team, cost protection, visibility	"DDoS Response Team" , "reimbursement for scaling costs during an attack" , "advanced DDoS"
GuardDuty	Threat detection from CloudTrail, VPC Flow Logs, and DNS logs, using machine learning	"detect unusual API calls ", "crypto-mining", "compromised instance", "no agents to install"
Inspector	Vulnerability scanning of EC2, ECR images, and Lambda	"scan for CVEs ", "software vulnerabilities", "unintended network exposure"

Service	What it does	Trigger phrase
Macie	Finds sensitive data in S3 using machine learning	" PII ", "credit card numbers in S3", "data classification"
Detective	Investigates the root cause of a finding	" investigate ", "analyse the cause of a security finding"
Security Hub	Aggregates findings from GuardDuty, Inspector, Macie, Config, and partners, and runs standards checks	" single pane of glass ", "CIS benchmark compliance score"
Audit Manager	Collects evidence for audits continuously	"audit evidence", "SOC 2 / PCI / HIPAA report preparation"
Artifact	Downloads AWS's own compliance reports and agreements	"I need AWS's SOC report", "sign a BAA"
Firewall Manager	Centrally applies WAF, Shield, security group, and Network Firewall policies across the organization	"apply the same rules to all accounts "
Network Firewall	Managed stateful firewall at the VPC level, with intrusion prevention	"deep packet inspection", "domain filtering for outbound VPC traffic"

The four that get confused. GuardDuty **detects** threats from logs. Inspector **scans** for vulnerabilities in software. Macie **classifies** sensitive data in S3. Detective **investigates** after the fact. Security Hub **collects** all of their findings.

Shared Responsibility Model

AWS is responsible for	You are responsible for
Security of the cloud	Security in the cloud
Hardware, the global infrastructure, regions, AZs, edge locations	Your data, its classification and encryption
The hypervisor and the managed service software	Guest OS patching on EC2 , application code
Physical and environmental controls	IAM users, roles, and policies
Decommissioning storage media	Security group and NACL configuration
Managed service patching (RDS, Lambda, DynamoDB)	Network traffic protection, client-side encryption

The line moves with the service. On **EC2** you patch the OS. On **RDS** AWS patches it. On **Lambda** and **S3** there is no OS to think about. Encryption and access control are **always** yours.

Organizations and multi-account

Cue	Answer
What Organizations gives you	consolidated billing, volume discounts pooled across accounts, SCPs , and account automation
Service Control Policy (SCP)	A guard rail that sets the maximum permissions for accounts in an OU. It never grants anything.
Does an SCP affect the management account	No — never. This is tested.
How to deny use of a region organization-wide	an SCP with a <code>aws:RequestedRegion</code> condition
Organizational unit (OU)	A folder of accounts you attach SCPs to
To spin up new accounts with guard rails already applied	AWS Control Tower
To deploy a stack to many accounts and regions at once	CloudFormation StackSets
To share a resource like a subnet or a Transit Gateway across accounts	AWS Resource Access Manager (RAM)
To stop spend surprises	AWS Budgets with alerts, plus Cost Anomaly Detection
To see cost by team	cost allocation tags plus Cost Explorer

Tagging

The four tag categories to name: **Technical, Automation, Business, and Security.**

Category	Examples
Technical	<code>Name</code> , <code>Application</code> , <code>Version</code> , <code>Cluster</code>
Automation	<code>Shutdown=22:00</code> , <code>Backup=daily</code> , <code>OptIn</code>
Business	<code>CostCenter</code> , <code>Owner</code> , <code>Project</code> , <code>BusinessUnit</code>
Security	<code>Confidentiality</code> , <code>Compliance</code> , <code>DataClass</code>

Cue	Answer
To require tags on new resources	an IAM or SCP condition on <code>aws:RequestTag</code> , or Tag Policies in Organizations
To grant access based on tags	attribute-based access control (ABAC) , using <code>aws:ResourceTag</code> conditions
Why ABAC scales better than one policy per team	you add resources and people without writing new policies

Trusted Advisor

Five categories — "**CPFSS**": **C**ost Optimization, **P**erformance, **F**ault Tolerance, **S**ecurity, **S**ervice Limits.

Cue	Answer
Which checks are free on Basic and Developer support	the core security checks and service limits
Which plans unlock all checks	Business, Enterprise On-Ramp, and Enterprise
Classic findings	idle load balancers, underutilized EC2 instances, unassociated Elastic IPs, S3 buckets open to the world, root account without MFA, exposed access keys, approaching a service limit

08 — Application Integration and Analytics

Pick the integration service

Shape of the need	Service
One producer, one consumer, work queued up, decoupling	SQS
One message, many subscribers, fan-out	SNS
Event-driven routing with filtering on content, plus third-party SaaS events	EventBridge
Multi-step workflow with branching, retries, and waiting	Step Functions
Managed RabbitMQ or ActiveMQ because the app already speaks AMQP, MQTT, STOMP, or JMS	Amazon MQ
Streaming data, many consumers, replayable, ordered per shard	Kinesis Data Streams
Managed Kafka	Amazon MSK

SQS — Simple Queue Service

Cue	Answer
What it buys you	loose coupling and elasticity — the producer and consumer never need to be up at the same time
Message size	256 KB , or up to 2 GB with the Extended Client Library plus S3
Standard queue guarantees	at-least-once delivery, best-effort ordering , nearly unlimited throughput
What that means for your code	design for duplicates and out-of-order messages — make processing safe to repeat
FIFO queue guarantees	exactly-once processing and strict order , 300 messages/s (3,000 batched, far more in high-throughput mode)
FIFO queue name must end in	.fifo
FIFO ordering key	the message group ID
FIFO deduplication	the message deduplication ID , or content-based deduplication, over a 5-minute window
Visibility timeout	how long a message is hidden after a receive. Default 30 seconds , max 12 hours .
Symptom: a message is processed twice	the visibility timeout is shorter than the processing time — raise it, or call ChangeMessageVisibility while working

Cue	Answer
Short polling versus long polling	Long polling waits up to 20 seconds for a message. It reduces empty responses and cost. Always prefer long polling.
Dead-letter queue	where messages go after maxReceiveCount failed attempts — use it to isolate poison messages
Delay queue	delays every message up to 15 minutes ; DelaySeconds delays an individual message
To scale consumers on queue depth	an Auto Scaling target tracking policy on ApproximateNumberOfMessagesVisible , or the backlog-per-instance metric
Encryption	SSE with KMS , plus TLS in transit
Access control	an SQS queue policy (resource-based)

The single most common SQS exam scenario: traffic spikes overwhelm the processing tier. Answer: **put an SQS queue between the front end and the workers, and auto-scale the workers on queue depth.**

SNS — Simple Notification Service

Cue	Answer
Model	publish/subscribe — one message to many subscribers
Subscriber (endpoint) types	HTTP/HTTPS, Email, Email-JSON, SQS, Lambda, SMS, mobile push, Kinesis Data Firehose
Topic types	standard and FIFO (FIFO topics can only deliver to SQS FIFO queues)
To send one message to several queues at once	fan-out — SNS topic with multiple SQS subscriptions
To have each subscriber receive only the messages it cares about	a subscription filter policy on message attributes
To capture messages a subscriber failed to receive	a redrive policy to a dead-letter queue
Message size	256 KB
Why fan-out beats having the producer write to each queue	the producer stays unaware of consumers, and you add subscribers without changing code

EventBridge

Cue	Answer
What it is	a serverless event bus that routes events by rules matching the event content
Event sources	AWS services , your own applications (custom bus), and SaaS partners (Zendesk, Datadog, Shopify)

Cue	Answer
Why pick it over SNS	content-based filtering on the whole event body , a schema registry , event replay and archive , and SaaS integration
Scheduled jobs	EventBridge Scheduler — cron or rate expressions, replacing CloudWatch Events rules
To react to an AWS API call	a rule matching a CloudTrail event
EventBridge Pipes	point-to-point: a source, optional filter and enrichment, a target — replaces a lot of glue Lambda code

SNS versus EventBridge in one line: SNS is for **high-throughput notification fan-out to known subscribers**. EventBridge is for **routing and filtering events across services and SaaS**, with replay.

Step Functions and SWF

Cue	Answer
Step Functions	Visual state machine that orchestrates Lambda, ECS, Batch, SNS, SQS, and over 200 services. Handles retries, error handling, parallel branches, choices, and waits .
Workflow types	Standard (up to 1 year, exactly-once, auditable) and Express (up to 5 minutes, very high volume, at-least-once)
When Step Functions is the answer	" coordinate multiple Lambda functions ", "a long-running process with human approval", "we need retry and error handling without writing it ourselves", "the workflow takes longer than 15 minutes"
SWF	The legacy workflow service. Only pick it if the question mentions external signals from humans or on-premises workers and explicitly rules out Step Functions. For new work, Step Functions.

API Gateway

Cue	Answer
What it fronts	Lambda, HTTP endpoints, any AWS service , or a VPC resource through a private integration
API types	REST (full features), HTTP (cheaper, faster, fewer features), WebSocket (two-way)
Endpoint types	Edge-optimized, Regional, Private (accessible only through a VPC interface endpoint)
To control who calls it	IAM authorization, Lambda authorizers, Cognito user pools , or resource policies
What an API key plus usage plan is for	throttling and metering per client — it is not authentication
To reduce backend load for repeated reads	API Gateway caching on a stage

Cue	Answer
To protect the backend from spikes	throttling — a steady-state rate and a burst limit
To return a 429	the client exceeded the throttle limit
To put a WAF in front	attach a web ACL to the regional stage

Amazon MQ

Pick it when the question says the application **already uses an industry-standard broker protocol** — **AMQP, MQTT, STOMP, OpenWire, or JMS** — and they want a lift-and-shift. For anything new and cloud-native, **SQS and SNS**.

Kinesis family

Service	What it does	Trigger
Data Streams	Ingests and retains a stream you write your own consumers for	"real time", "replay" , "multiple applications read the same stream", "ordered per key", "clickstream", "IoT telemetry", "financial transactions", "log and location-tracking events"
Data Firehose	Delivers a stream into a destination with no code	"load into S3 / Redshift / OpenSearch / Splunk" , "near real time is fine", "no servers to manage", "transform with Lambda on the way"
Managed Service for Apache Flink	SQL or Flink over a live stream	"analyse the stream with SQL" , "windowed aggregation in real time", "anomaly detection on the stream"
Video Streams	Ingest and index video	"camera feeds", "video for ML"

Cue	Answer
Unit of scale in Data Streams	the shard
Shard write and read capacity	1 MB/s or 1,000 records/s in; 2 MB/s out (shared)
To give each consumer its own 2 MB/s	enhanced fan-out
Record retention	24 hours default, up to 365 days
Capacity modes	provisioned (you set shards) and on-demand (AWS scales it)
Symptom: ProvisionedThroughputExceededException	too few shards, or a hot partition key — use a higher-cardinality partition key, or add shards
Kinesis versus SQS	Kinesis keeps the data for replay and lets many consumers read the same records in order . SQS deletes a message once it is processed, and each message goes to one consumer.
Firehose latency floor	about 60 seconds — so Firehose is not the answer to "true real time"

Analytics

Service	One-line trigger
Athena	" SQL directly on S3 ", "no infrastructure", "pay per query, per TB scanned", "ad-hoc analysis of logs in S3"
Glue	" serverless ETL ", " Data Catalog ", "crawl S3 and infer the schema", "prepare data for Athena or Redshift"
EMR	" Hadoop, Spark, Hive, HBase, Presto ", "I need access to the cluster and the file system", "customize the framework". EMR launches EC2 instances you can see and administer .
Redshift	" data warehouse ", "petabyte scale", "complex joins and high-concurrency BI"
OpenSearch Service	" full-text search ", "log analytics dashboards", "search an index"
QuickSight	" BI dashboards ", "visualize for business users", "embedded analytics"
Lake Formation	" build a data lake with fine-grained, column-level permissions in one place"
Data Pipeline	legacy scheduled data movement — prefer Glue or Step Functions for new work
DataZone	data catalog and governance for sharing data across teams

■ Cost-saving tricks for Athena the exam likes

- Store data in a **columnar format** — **Parquet or ORC** rather than CSV or JSON.
- **Compress** the files.
- **Partition** by the columns you filter on, such as year and month.
- Use **fewer, larger files** instead of many tiny ones.

All four reduce **bytes scanned**, which is exactly what Athena charges for.

SES — Simple Email Service

Cue	Answer
What it is for	sending and receiving bulk or transactional email
SES versus SNS for email	SES for real email with templates, attachments, and deliverability reporting. SNS email subscriptions for simple notifications.
Before you can send at volume	verify your domain or email address and request removal from the sandbox
Deliverability features	DKIM, SPF, dedicated IPs, configuration sets, reputation dashboard

Media, machine learning, and the rest you should recognize

Service	One-line trigger
AppSync	managed GraphQL API, with offline sync for mobile

Service	One-line trigger
Cognito user pools	sign-up and sign-in for your app's users
Cognito identity pools	exchange a login for temporary AWS credentials
Amplify	full-stack hosting and backend for web and mobile front ends
Elastic Transcoder / MediaConvert	convert video formats
Rekognition	images and video — faces, objects, moderation
Comprehend	natural language — sentiment, entities, PII detection in text
Textract	extract text, forms, and tables from scanned documents
Transcribe	speech to text
Polly	text to speech
Translate	language translation
Forecast	time-series forecasting
Personalize	recommendations
Fraud Detector	online fraud detection
SageMaker	build, train, and deploy your own models
Bedrock	managed access to foundation models
IoT Core	connect and manage devices at scale, MQTT
Device Farm	test apps on real phones
WorkSpaces	managed virtual desktops
AppStream 2.0	stream a single application to a browser

09 — Monitoring, Management, and Governance

The three that always get confused

Service	The one-word answer	Question
CloudWatch	Performance	"How is it behaving? What are the metrics and logs?"
CloudTrail	Audit	" Who did what , when , and from which IP ?"
AWS Config	Compliance	"What is the configuration now, what was it before, and does it meet our rules?"

Memorize this sentence: "**CloudWatch watches performance. CloudTrail trails API calls. Config tracks configuration.**"

CloudWatch

Cue	Answer
What it collects	metrics, logs, events, and traces (with X-Ray)
Basic versus detailed monitoring	5 minutes free, 1 minute paid
High-resolution metrics	down to 1 second
Metric retention	1-second data 3 hours ; 1-minute 15 days ; 5-minute 63 days ; 1-hour 455 days
Log retention	1 day to 10 years, or never expire
EC2 metrics available with no agent	CPUUtilization, NetworkIn/Out, DiskReadOps/WriteOps, StatusCheckFailed
Metrics that require the CloudWatch agent	memory used, disk space used, swap usage, per-process metrics
What the agent is called now	the unified CloudWatch agent (it replaced the old CloudWatch Logs agent)
Alarm states	OK, ALARM, INSUFFICIENT_DATA
What an alarm can do	notify SNS , trigger Auto Scaling , run an EC2 action (stop, terminate, reboot, recover), or start a Systems Manager action
Composite alarm	combines several alarms with AND/OR to cut noise
To search and analyse log data with a query language	CloudWatch Logs Insights
To turn a log pattern into a metric	a metric filter — for example, count ERROR lines and alarm on them
To stream logs onward	a subscription filter to Lambda, Kinesis, or Firehose
To archive logs cheaply for years	export to S3 , then lifecycle them to Glacier
To see all your dashboards from several accounts	cross-account cross-region dashboards

Cue	Answer
To watch an endpoint from outside	CloudWatch Synthetics canaries
To see real user browser performance	CloudWatch RUM
To watch container workloads	Container Insights
To get recommendations on anomalies automatically	CloudWatch Anomaly Detection
Route 53 health checks integrate with	CloudWatch — and a CloudWatch alarm can itself be a health check

CloudTrail

Cue	Answer
What it records	API activity across the console, SDKs, CLI, and other AWS services — governance, compliance, operational and risk auditing
What is on by default	90 days of Event history in every region, for management events, at no cost
What you must create for anything longer	a trail delivering to S3 (and optionally CloudWatch Logs)
Best practice when creating a trail	apply it to all regions , so future regions are included automatically
Event types	management events (control plane, on by default), data events (S3 object-level and Lambda invocations, off by default and charged), Insights events (unusual activity)
To know who deleted an S3 object	enable data events for that bucket — management events alone will not show it
Encryption	log files are encrypted with SSE-S3 by default ; you can use SSE-KMS
To prove logs have not been tampered with	log file integrity validation , which produces a digest file
To collect trails from every account in one place	an organization trail , delivering to a central S3 bucket
To search trail data with SQL	Athena over the S3 bucket
Delivery delay	typically within 15 minutes of the API call

AWS Config

Cue	Answer
What it does	records the configuration of your resources over time and evaluates them against rules
What a configuration item is	a point-in-time snapshot of a resource's attributes and relationships
Managed rules you should recognize	"S3 bucket must not be public" , "EBS volumes must be encrypted" , "root account must have MFA" , "required tags must be present" , "security groups must not allow unrestricted SSH"
Custom rules run on	Lambda or Guard
To fix a non-compliant resource automatically	a remediation action , usually a Systems Manager Automation document

Cue	Answer
To apply rules across every account	Config conformance packs and aggregators
What it answers that CloudTrail cannot	" what did this security group look like last Tuesday? " and " is this resource compliant right now? "

CloudFormation

Cue	Answer
What it is	infrastructure as code in JSON or YAML, creating a stack of resources
Template sections	AWS TemplateFormatVersion, Description , Metadata , Parameters , Mappings , Conditions , Transform , Resources , Outputs
The only required section	Resources
Parameters	Values passed in at deploy time
Mappings	Fixed lookup tables, classically region to AMI ID
Outputs	Values exported from the stack, importable by another stack
Conditions	Create a resource only when a test passes, for example only in production
To preview what a change will do	a change set
To stop an update deleting a resource	a stack policy , or DeletionPolicy: Retain
To keep data when the stack is deleted	DeletionPolicy: Snapshot or Retain
To find out whether someone changed a resource outside the template	drift detection
To deploy the same stack across many accounts and regions	StackSets
To reuse a template inside another	nested stacks
To signal that an EC2 instance finished bootstrapping	a CreationPolicy with cfn-signal and a WaitCondition
To define resources in a real programming language	the AWS CDK , which synthesizes CloudFormation
To offer approved templates to teams in a self-service catalog	AWS Service Catalog
What happens on a failed create	automatic rollback by default

Beanstalk, CloudFormation, and OpsWorks in one line each. Beanstalk: "deploy my app, you decide the infrastructure." CloudFormation: "build exactly the infrastructure I declared." OpsWorks: "we already use Chef or Puppet, give us managed Chef or Puppet servers."

Systems Manager (SSM)

Capability	What it does
Session Manager	Browser or CLI shell to an instance with no open inbound ports, no SSH keys, no bastion — fully logged
Run Command	Run a script or command across many instances at once
Patch Manager	Scan and install OS patches on a schedule, with baselines
Parameter Store	Free hierarchical config store, with SecureString for encrypted values
State Manager	Keep instances in a defined state
Automation	Runbooks for repetitive operations, and Config remediation
Inventory	What software is installed where
Compliance	Patch and association compliance reporting
Fleet Manager	Manage instances without logging in
Maintenance Windows	Scheduled windows for disruptive tasks

The trigger to remember: "we must not open port 22 and must not manage SSH keys, but engineers need shell access, and it must be auditable" → **Session Manager**.

Cost management

Service	Trigger
Cost Explorer	"visualize and forecast spend", "break down by service, tag, or account"
AWS Budgets	" alert me when spend or usage crosses a threshold", "budget for a Reserved Instance utilization target"
Cost and Usage Report (CUR)	"the most detailed line-item billing data, delivered to S3"
Cost Anomaly Detection	"detect unexpected spend automatically with machine learning"
Compute Optimizer	" right-size my EC2, EBS, Lambda, and ECS resources based on real utilization"
Trusted Advisor	"idle resources, unassociated Elastic IPs, approaching service limits"
Pricing Calculator	"estimate the cost before we build it"
Billing Conductor	"show different rates to different internal business units"

■ The classic cost-optimization answers

Problem	Answer
Instances idle at night and weekends	scheduled Auto Scaling , or an Instance Scheduler on tags
Steady 24/7 workload on On-Demand	Savings Plans or Reserved Instances

Problem	Answer
Batch jobs that can be interrupted	Spot Instances
Large NAT gateway data processing bill for S3 traffic	an S3 gateway VPC endpoint
Storing everything in S3 Standard forever	lifecycle rules to IA and Glacier, and Intelligent-Tiering when the pattern is unknown
Paying for old snapshots and unattached volumes	find them with Trusted Advisor or Config , then delete, and automate retention with DLM or AWS Backup
High Athena bill	Parquet, compression, and partitioning
Over-sized instances	Compute Optimizer
Cross-AZ data transfer charges	keep chatty tiers in the same AZ , while still spanning AZs for availability
Over-provisioned DynamoDB	on-demand mode for spiky traffic, or auto scaling on provisioned

Service Health and support

Cue	Answer
AWS Health Dashboard	Events that affect your resources — scheduled maintenance, degraded services
Service Quotas	View and request increases to limits, in one place
Support plans	Basic (free), Developer, Business, Enterprise On-Ramp, Enterprise
Which plan gives you a Technical Account Manager	Enterprise (and a pooled TAM on Enterprise On-Ramp)
Which plans give full Trusted Advisor	Business and above
Which plan is the minimum for 24/7 phone and chat with engineers	Business

Regions, Availability Zones, and resilience

Cue	Answer
Region	A geographic area with at least 3 (usually 3 or more) Availability Zones
Availability Zone	One or more discrete datacentres with independent power, cooling, and networking, tens of kilometres apart
Local Zone	An extension of a region placed near a large city for single-digit-millisecond latency
Wavelength Zone	Inside a telecom provider's 5G network, for mobile edge latency
Outposts	AWS racks in your own datacentre , same APIs — for data residency or very low on-premises latency
Edge location	A CloudFront point of presence — far more numerous than regions
How to survive an AZ failure	spread across at least 2 AZs , usually 3

Cue	Answer
How to survive a region failure	multi-region — cross-region replication, Route 53 failover, Aurora Global Database, DynamoDB Global Tables
Which services are global	IAM, Route 53, CloudFront, WAF (for CloudFront), Organizations, Billing
Which services are regional	nearly everything else — EC2, S3 buckets, RDS, VPC, DynamoDB tables
Which resources are AZ-scoped	subnets, EBS volumes, EC2 instances, ENIs, NAT gateways, RDS instances
Things that must be copied to another region	AMIs, EBS snapshots, RDS snapshots, ECR images, Redshift snapshots, KMS keys (or use multi-Region keys)

The resilience question in one sentence: if the question says "**highly available**", the answer involves **multiple Availability Zones**. If it says "**disaster recovery**" or "**region outage**", the answer involves **multiple regions**.

10 — Decision Triggers

Most of the exam is scenario questions. This page is the scenario-to-service map. **If you only drill one page, drill this one.**

How to use it: cover the right column. Read the left. Say the service. The exam phrases things almost exactly like this.

Keyword reflexes

These words, on their own, should produce an instant answer.

The question says	Reach for
"decouple"	SQS
"fan-out", "notify many subscribers"	SNS
"event-driven", "filter on event content", "SaaS event"	EventBridge
"orchestrate", "coordinate steps", "retry and branch"	Step Functions
"serverless"	Lambda, API Gateway, DynamoDB, S3, Fargate, Aurora Serverless
"highly available"	multiple Availability Zones
"disaster recovery", "region outage"	multiple regions
"durable"	S3, or an RDS read replica / Aurora's 6 copies
"in-memory", "sub-millisecond", "microsecond"	ElastiCache, or DAX for DynamoDB
"petabyte-scale analytics", "data warehouse"	Redshift
"SQL on data in S3", "no infrastructure"	Athena
"Hadoop", "Spark", "Hive", "HBase", "Presto"	EMR
"full-text search", "log dashboards"	OpenSearch Service
"CDN", "cache at the edge", "global users, static content"	CloudFront
"static IP addresses", "TCP/UDP", "gaming", "VoIP"	Global Accelerator
"DNS", "route traffic by region or weight"	Route 53
"dedicated private network connection"	Direct Connect
"encrypted tunnel over the internet, quickly"	Site-to-Site VPN
"connect hundreds of VPCs"	Transit Gateway
"connect two VPCs"	VPC peering
"private access to S3 from a private subnet"	S3 gateway VPC endpoint
"private access to another AWS service or a partner service"	interface VPC endpoint (PrivateLink)

The question says	Reach for
"no bandwidth, petabytes, deadline"	Snowball / Snowmobile
"ongoing incremental sync from on-premises"	DataSync
"on-premises app needs cloud storage behind it"	Storage Gateway
"shared file system, Linux, many instances"	EFS
"SMB", "Windows file share", "Active Directory"	FSx for Windows File Server
"HPC shared scratch", "hundreds of GB/s"	FSx for Lustre
"infrastructure as code", "repeatable environments"	CloudFormation
"upload my code and let AWS handle the rest"	Elastic Beanstalk
"Chef or Puppet"	OpsWorks
"Kubernetes", "kubect!"	EKS
"containers with no servers to manage"	Fargate
"who made this API call?"	CloudTrail
"what was this configuration last week?", "is it compliant?"	AWS Config
"metrics, alarms, logs"	CloudWatch
"shell access with no open ports and no SSH keys"	Session Manager
"rotate database credentials automatically"	Secrets Manager
"free config store"	Parameter Store
"SQL injection", "cross-site scripting", "rate limit by IP"	AWS WAF
"DDoS Response Team", "cost reimbursement during an attack"	Shield Advanced
"detect unusual API activity, no agents"	GuardDuty
"scan for CVEs and software vulnerabilities"	Inspector
"find PII in S3"	Macie
"single pane of glass for security findings"	Security Hub
"single-tenant HSM", "FIPS 140-2 Level 3", "we must be the only key holders"	CloudHSM
"guard rails across all accounts"	Organizations SCPs (or Control Tower to set it all up)
"right-size my resources based on real usage"	Compute Optimizer
"alert me when spend exceeds a threshold"	AWS Budgets
"immutable, cryptographically verifiable history"	QLDB
"multiple parties, no central authority"	Managed Blockchain

The question says	Reach for
"graph", "relationships", "fraud ring"	Neptune
"time series", "IoT sensor readings over time"	Timestream
"virtual desktops"	WorkSpaces
"stream one application to a browser"	AppStream 2.0
"data residency", "AWS hardware in my datacentre"	Outposts
"single-digit millisecond latency to a specific city"	Local Zones
"5G mobile edge"	Wavelength

Compute scenarios

Scenario	Answer
Steady, predictable 24/7 workload, lowest cost	Savings Plans or Reserved Instances
Batch processing that can be interrupted and restarted	Spot Instances
Spot plus guaranteed baseline capacity	a mixed-instances Auto Scaling group: On-Demand or RI baseline plus Spot for the rest
Software licensed per physical socket or core	Dedicated Host
Must be certain capacity exists in an AZ for a launch next week, no commitment	On-Demand Capacity Reservation
Instances need the lowest possible network latency between them	cluster placement group
A handful of critical instances must never share hardware	spread placement group
Large distributed system like Cassandra or HDFS	partition placement group
A function needs to run for 40 minutes	not Lambda — use Fargate, ECS, or AWS Batch
A Lambda function has slow cold starts and latency matters	provisioned concurrency
A Lambda function is too slow and the code cannot change	increase its memory , which also increases CPU
One function is consuming the whole account's concurrency	reserved concurrency on that function
Instances are terminated and relaunched repeatedly by Auto Scaling	increase the health check grace period ; point the ELB health check at a real readiness endpoint
Auto Scaling keeps scaling in and out rapidly	increase the cooldown and widen the alarm threshold
Need to scale for a known Monday 9am spike	scheduled scaling
Need to scale on a learned daily pattern	predictive scaling

Scenario	Answer
Need to keep average CPU at a target	target tracking
Must run a cleanup script before an instance is terminated	a lifecycle hook
Containers need GPUs or a specific instance type	ECS or EKS on the EC2 launch type , not Fargate
Want the least operational overhead for containers	Fargate
Need the same image in two regions	replicate the ECR repository — images are region-scoped

Storage scenarios

Scenario	Answer
Objects must survive the loss of an Availability Zone	S3 Standard, Standard-IA, or any Glacier class — not One Zone-IA
Thumbnails that can be regenerated, cheapest storage	S3 One Zone-IA
Access pattern is unknown or changes over time	S3 Intelligent-Tiering
Compliance archive, retrieval in 12 to 48 hours is fine, cheapest possible	Glacier Deep Archive
Archive that must still be readable in milliseconds	Glacier Instant Retrieval
Records must not be deletable for seven years, even by an administrator	S3 Object Lock in compliance mode (plus versioning)
Protect against accidental overwrite and delete	versioning , and MFA Delete for the strong version
Must survive a region outage	Cross-Region Replication
Keep a second copy in the same region for log aggregation or compliance	Same-Region Replication
Replication is configured but old objects did not copy	S3 Batch Replication — replication only applies to new objects
Users worldwide upload large files slowly	S3 Transfer Acceleration
Uploading a 20 GB file	multipart upload — required above 5 GB
Let an unauthenticated browser upload one file securely	a presigned URL
Serve a private bucket through CloudFront	Origin Access Control
Bucket is public and should not be	S3 Block Public Access
Need to delete 50 million objects older than a year	lifecycle expiration rule
Incomplete multipart uploads are costing money	lifecycle rule to abort incomplete multipart uploads
A boot volume that is cheap and fast enough for most workloads	gp3
A database needing guaranteed 40,000 IOPS	io2

Scenario	Answer
Sequential throughput for log processing	st1
Need the same volume attached to two instances	io1 or io2 with Multi-Attach , same AZ — or rethink and use EFS
Move an EBS volume to another AZ	snapshot, then create a volume from the snapshot in the target AZ
Encrypt an existing unencrypted EBS volume or RDS instance	snapshot it, copy the snapshot with encryption enabled, restore from the copy
Highest possible IOPS and the data is disposable	instance store
Backups of many services under one policy, with cross-region copies	AWS Backup
Retire an on-premises tape library	Tape Gateway
Shrink the on-premises storage footprint but keep low-latency access to hot data	Volume Gateway, cached mode
Keep the whole dataset on-premises with async backup to AWS	Volume Gateway, stored mode
On-premises apps need an NFS or SMB share backed by S3	S3 File Gateway

Database scenarios

Scenario	Answer
RDS must survive an AZ failure with automatic failover	Multi-AZ
Reporting queries are slowing the production database	a read replica (and point the reports at it)
Reads must scale and the same queries repeat constantly	ElastiCache in front, or a read replica
Need a read replica in another region for both DR and local reads	cross-region read replica
MySQL-compatible, needs far more performance and 15 replicas	Aurora
Cross-region DR with sub-second lag and about a minute to fail over	Aurora Global Database
Traffic is intermittent, want to pay nothing much when idle	Aurora Serverless v2
Need a test copy of a 2 TB Aurora database in minutes	Aurora fast database cloning
Need to undo a bad update on Aurora MySQL in place	Backtrack
Needs OS-level access or an engine RDS does not support	run the database on EC2
Single-digit millisecond key-value at any scale, no servers	DynamoDB

Scenario	Answer
DynamoDB reads must be microseconds	DAX
Need an active-active DynamoDB table in three regions	Global Tables
Need to react to every item change in DynamoDB	DynamoDB Streams plus Lambda
Need to purge old DynamoDB items at no cost	TTL
DynamoDB throttling with an uneven access pattern	redesign the partition key for higher cardinality , or switch to on-demand
Unpredictable, spiky DynamoDB traffic	on-demand capacity mode
Need session storage that survives a node failure	ElastiCache for Redis with Multi-AZ, or DynamoDB
Simplest multi-threaded cache, no persistence needed	ElastiCache for Memcached
Need a leaderboard or sorted set	ElastiCache for Redis
Complex BI reporting across billions of rows, high concurrency	Redshift
Occasional ad-hoc SQL over logs already in S3	Athena
Query S3 data from an existing Redshift cluster without loading it	Redshift Spectrum
Short queries are stuck behind long ones in Redshift	Workload Management (WLM) query queues
Migrate Oracle to Aurora PostgreSQL with minimal downtime	Schema Conversion Tool plus DMS

Networking scenarios

Scenario	Answer
Private instances need to download OS updates	NAT gateway in a public subnet , with a route from the private subnet
Private instances need S3 and nothing else	S3 gateway endpoint — cheaper than a NAT gateway
Block one malicious IP address	a NACL deny rule — security groups cannot deny
Allow the web tier to reach the app tier	a security group rule on the app tier referencing the web tier's security group
SSH works outbound but not inbound through a NACL	add the inbound rule and the outbound ephemeral port range (1024 to 65535) — NACLs are stateless
Instance in a public subnet with a public IP still cannot reach the internet	check the route table has 0.0.0.0/0 to the internet gateway , then the security group, then the NACL
Need to route <code>/api</code> and <code>/static</code> to different target groups	ALB path-based routing
Need a load balancer with a fixed IP for a firewall allow-list	NLB
Need UDP load balancing	NLB

Scenario	Answer
Need millions of requests per second with the lowest latency	NLB
Need to redirect all HTTP to HTTPS without touching the app	an ALB listener rule
A third-party security appliance must inspect all VPC traffic	Gateway Load Balancer
Three VPCs must all reach each other	Transit Gateway — peering is not transitive
Need to expose an internal service to a customer's VPC without peering	PrivateLink with an NLB
Hybrid link needed within days	Site-to-Site VPN
Hybrid link needing consistent high bandwidth, cost is fine	Direct Connect
Highest-resilience hybrid connectivity	two Direct Connect connections at two locations, with VPN backup
Resolve on-premises hostnames from inside a VPC	Route 53 Resolver outbound endpoint
Fail over to a static maintenance page during an outage	Route 53 failover routing to an S3 static website
EU users must be served from the EU for data residency	geolocation routing
Serve every user from whichever region is fastest for them	latency routing
Shift 5% of traffic to a new stack	weighted routing
Reduce latency for a global audience reading the same files	CloudFront
Reduce latency for a global audience on a TCP game protocol	Global Accelerator
Restrict video to subscribers	CloudFront signed URLs or signed cookies
Block a whole country from a CloudFront distribution	geo-restriction
Rewrite a request header at the edge, very cheaply	CloudFront Functions
Need to call DynamoDB from edge logic	Lambda@Edge
Find out why traffic is being dropped in a VPC	VPC Flow Logs — look for REJECT
Need to capture actual packet contents	VPC Traffic Mirroring

Security scenarios

Scenario	Answer
An application on EC2 needs to read an S3 bucket	an IAM role attached via an instance profile — never access keys in code

Scenario	Answer
A partner's AWS account needs limited access	a cross-account IAM role with a trust policy , assumed with STS
Employees should sign in once for all accounts	IAM Identity Center
Mobile app users should get temporary AWS credentials	Cognito identity pools
Nobody in any account may use a region	an SCP with an <code>aws:RequestedRegion</code> condition
Developers must not be able to escalate their own permissions	a permissions boundary
Database password must rotate every 30 days with no code change	Secrets Manager with rotation
Need a free place to store non-secret configuration	Parameter Store
Must prove no one tampered with the audit log	CloudTrail log file integrity validation , and S3 Object Lock on the log bucket
Need to know who deleted an object from a bucket	CloudTrail data events for that bucket
Need to alarm when the root account is used	a CloudWatch Logs metric filter on the CloudTrail log group , plus an SNS alarm
Must detect compromised instances mining cryptocurrency	GuardDuty
Must report on unpatched software across the fleet	Inspector (findings) and Systems Manager Patch Manager (fixing)
Must find credit card numbers accidentally stored in S3	Macie
Must stop an application-layer attack on a public web app	AWS WAF
Must satisfy an auditor asking for AWS's SOC 2 report	AWS Artifact
Must encrypt data where AWS can never access the keys	CloudHSM , or SSE-C for S3
Must enforce encryption on every new EBS volume	account-level EBS encryption by default , plus an AWS Config rule to catch exceptions
Must enforce that no S3 bucket is public, across the organization	S3 Block Public Access at the account level , an SCP , and a Config rule
Engineers need audited shell access with no open ports	Session Manager

The tie-breakers

When two answers both technically work, apply these in order:

1. **Managed beats self-managed.** Fargate over EC2 containers. RDS over a database on EC2. Prefer the answer with less for you to patch.
2. **Multi-AZ beats single-AZ** whenever the question mentions availability.
3. **Temporary credentials beat long-term keys.** A role always beats an access key.
4. **An explicit deny or a guard rail beats trusting people.** SCP, boundary, Block Public Access.

5. **Cheaper wins only if all stated requirements are still met.** Spot is wrong the moment the question says "cannot be interrupted."
6. **Serverless wins on "minimal operational overhead."** That phrase is a near-certain signal.
7. **Reject anything retired.** EC2-Classic, Spot block, Classic Load Balancer for new designs, Data Pipeline for new ETL, SWF for new workflows.
8. **Reject anything physically impossible.** A subnet spanning AZs. An EBS volume across AZs. An Alias record to an EC2 instance name. A CNAME at the zone apex. Reading from a Multi-AZ standby.

11 — Rapid Fire

150 one-liners. Cover the right column, work down the page, and mark the ones you miss. About 12 minutes end to end. Do it every morning in the last two weeks before the exam.

If you miss three in the same group, go back and reread that service page.

Foundations (1–15)

#	Cue	Answer
1	A region contains at least how many AZs	3 (some older ones have 2; design for 3)
2	Can a subnet span two AZs	No
3	Can a VPC span two regions	No
4	Name the global services	IAM, Route 53, CloudFront, Organizations, WAF for CloudFront, Billing
5	Which resources are AZ-scoped	subnets, EC2 instances, EBS volumes, ENIs, NAT gateways
6	Which artifacts must be copied to another region	AMIs, EBS snapshots, RDS snapshots, ECR images, KMS keys
7	The six Well-Architected pillars	Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability
8	AWS is responsible for security of what	the cloud — hardware, infrastructure, managed service software
9	You are responsible for security in what	the cloud — your data, IAM, guest OS on EC2, security groups
10	Who patches the OS on RDS	AWS
11	Who patches the OS on EC2	you
12	RTO means	Recovery Time Objective — how long you may be down
13	RPO means	Recovery Point Objective — how much data you may lose
14	The four DR strategies, cheapest first	backup and restore, pilot light, warm standby, multi-site active-active
15	The 7 Rs of migration	retire, retain, relocate, rehost, replatform, refactor, repurchase

EC2 and Auto Scaling (16–40)

#	Cue	Answer
16	The seven instance states	pending, running, stopping, stopped, shutting-down, terminated, rebooting
17	What survives a reboot that does not survive a stop	instance store data and the public IPv4 address
18	What hibernate saves	the contents of RAM, written to the root EBS volume
19	Metadata endpoint	169.254.169.254
20	Which metadata service version to prefer	IMDSv2
21	When does user data run	once, at first boot, as root
22	Spot interruption notice	2 minutes
23	Spot discount	up to about 90%
24	RI and Savings Plan discount	up to about 72%
25	Needed for a socket-based software licence	Dedicated Host
26	Lowest-latency placement	cluster placement group
27	Maximum isolation placement	spread placement group
28	Placement group for Cassandra or HDFS	partition
29	Default user for an Amazon Linux AMI	ec2-user
30	Default user for an Ubuntu AMI	ubuntu
31	System status check failing means	AWS's infrastructure — stop and start to migrate hosts
32	Instance status check failing means	your OS or configuration
33	Default Auto Scaling cooldown	300 seconds
34	Default health check grace period	300 seconds
35	Simplest recommended scaling policy	target tracking
36	Policy for a known Monday-morning spike	scheduled scaling

#	Cue	Answer
37	Legacy versus current instance definition	launch configurations are legacy; launch templates are current and versioned
38	To take an instance out of service for debugging	standby
39	To run a script before termination	a lifecycle hook
40	Besides EC2, name three things Auto Scaling can scale	ECS service count, DynamoDB capacity, Aurora replica count, Spot Fleet

Load balancing (41–52)

#	Cue	Answer
41	ALB layer	7
42	NLB layer	4
43	Gateway Load Balancer layer	3
44	Which load balancer gives you a static IP	NLB
45	Which supports UDP	NLB
46	Which does path and host routing	ALB
47	Which can authenticate with Cognito	ALB
48	Minimum AZs for an internet-facing load balancer	2
49	Cross-zone load balancing default on ALB	on
50	Cross-zone load balancing default on NLB	off
51	Where the health check is configured	on the target group
52	How to serve multiple certificates on one listener	SNI

Lambda and serverless (53–64)

#	Cue	Answer
53	Lambda max duration	15 minutes (900 seconds)
54	Lambda memory range	128 MB to 10,240 MB
55	Lambda synchronous payload limit	6 MB
56	Lambda asynchronous payload limit	256 KB
57	Lambda <code>/tmp</code> default size	512 MB , configurable to 10,240 MB

#	Cue	Answer
58	Default concurrency per region	1,000
59	Fix cold starts	provisioned concurrency
60	Stop one function starving the account	reserved concurrency
61	Share dependencies between functions	layers
62	What a 429 from Lambda means	throttled — concurrency limit hit
63	API key plus usage plan is for	throttling and metering, not authentication
64	API Gateway options for auth	IAM, Lambda authorizer, Cognito user pool, resource policy

S3 (65–85)

#	Cue	Answer
65	Max object size	5 TB
66	Single PUT limit	5 GB
67	Multipart part size range	5 MB to 5 GB
68	Max parts	10,000
69	Durability	11 nines
70	Consistency model	strong read-after-write
71	Default encryption today	SSE-S3, applied automatically to new objects
72	SSE-C means	you supply the key with every request; AWS never stores it
73	SSE-KMS means	you control the customer master key; AWS handles the data key
74	Requirement for Cross-Region Replication	versioning on both buckets, plus an IAM role
75	Does replication copy existing objects	no — use S3 Batch Replication
76	Minimum days before a Standard-IA transition	30
77	Minimum storage duration for Glacier Deep Archive	180 days

#	Cue	Answer
78	Glacier Flexible Retrieval expedited time	1 to 5 minutes
79	Glacier Flexible Retrieval standard time	3 to 5 hours
80	Glacier Flexible Retrieval bulk time	5 to 12 hours
81	How long restored Glacier data stays available	24 hours
82	Which class is single-AZ	One Zone-IA
83	Requirement for MFA Delete	versioning, and only the root user can enable it, via the CLI
84	Make objects undeletable for seven years	Object Lock in compliance mode
85	Current name for CloudFront-to-private-bucket access	Origin Access Control (OAC)

EBS, EFS, FSx (86–97)

#	Cue	Answer
86	gp3 baseline	3,000 IOPS, 125 MB/s
87	gp2 IOPS per GB	3 , capped at 16,000
88	io2 max IOPS	64,000 (Block Express 256,000)
89	Max EBS volume size	16 TiB
90	Which types cannot be boot volumes	st1 and sc1
91	Which types support Multi-Attach	io1 and io2 , up to 16 instances, same AZ
92	Where snapshots live	S3, incrementally, region-scoped
93	Can you shrink an EBS volume	No
94	Encrypt an existing unencrypted volume	snapshot, copy with encryption on, restore
95	EFS protocol and OS	NFS v4.1, Linux only
96	EFS scope	regional, with mount targets per AZ
97	SMB and Active Directory file shares	FSx for Windows File Server

Databases (98–120)

#	Cue	Answer
98	Multi-AZ replication type	synchronous
99	Read replica replication type	asynchronous
100	Can you read from a Multi-AZ standby	No
101	Multi-AZ purpose	availability
102	Read replica purpose	read scaling
103	Read replicas per RDS instance	5
104	Read replicas per Aurora cluster	15
105	Default RDS backup retention	7 days , range 0 to 35
106	RDS point-in-time recovery granularity	about 5 minutes
107	Restoring an RDS snapshot gives you	a new instance with a new endpoint
108	Aurora copies of your data	6 across 3 AZs
109	Max Aurora cluster volume	128 TiB
110	Aurora cross-region DR feature	Global Database
111	Aurora feature to rewind in place	Backtrack
112	DynamoDB max item size	400 KB
113	What 1 RCU buys	one strongly consistent read/s of up to 4 KB
114	What 1 WCU buys	one write/s of up to 1 KB
115	DynamoDB microsecond cache	DAX
116	LSI must be created when	with the table
117	GSI consistency	eventually consistent only
118	DynamoDB cross-region active-active	Global Tables
119	Which ElastiCache engine supports persistence and failover	Redis
120	Which ElastiCache engine is multi-threaded	Memcached

Networking (121–138)

#	Cue	Answer
121	VPC CIDR range	/16 to /28

#	Cue	Answer
122	Reserved IPs per subnet	5
123	Usable IPs in a /24	251
124	What makes a subnet public	a route to an internet gateway
125	Security groups are	stateful, allow-only, instance level
126	NACLs are	stateless, allow and deny, subnet level, evaluated in order
127	How to block a single IP	a NACL deny rule
128	NACL rules needed for SSH	inbound 22 and outbound ephemeral ports 1024–65535
129	Default NACL behaviour	allows everything
130	Custom NACL behaviour	denies everything
131	Is VPC peering transitive	No
132	Can peered VPCs overlap CIDRs	No
133	Transitive hub for many VPCs	Transit Gateway
134	Which services use gateway endpoints	S3 and DynamoDB only
135	Alias versus CNAME at the apex	only Alias works at the zone apex
136	Route 53 multivalue returns	up to 8 healthy records
137	CloudFront certificate region	us-east-1
138	CloudFront default TTL	24 hours

Integration, monitoring, security (139–150)

#	Cue	Answer
139	SQS max message size	256 KB
140	SQS default and max retention	4 days default, 14 days max
141	SQS default visibility timeout and max	30 seconds, 12 hours
142	SQS long polling max wait	20 seconds
143	SQS standard delivery guarantee	at-least-once, best-effort ordering
144	FIFO queue name suffix	.fifo

#	Cue	Answer
145	Kinesis shard capacity in and out	1 MB/s in, 2 MB/s out
146	Kinesis default and max retention	24 hours default, 365 days max
147	CloudWatch basic versus detailed interval	5 minutes versus 1 minute
148	Metrics needing the CloudWatch agent	memory and disk space used
149	CloudTrail free event history	90 days, management events
150	Trusted Advisor's five categories	Cost Optimization, Performance, Fault Tolerance, Security, Service Limits

99 — Review of the Original Slide Decks

What I reviewed

Seven PowerPoint files in `G:\My Drive\Resources\Education\AWS\Architect Associate` :

File	Slides	Notes
<code>aws-architect-associate.pptx</code>	2	Earliest version
<code>aws-architect-associate-1.pptx</code>	2	Near-identical to the above
<code>aws-architect-associate-2.pptx</code>	3	Adds S3 Infrequent Access, EBS snapshot CLI
<code>aws-architect-associate-3.pptx</code>	7	Expands
<code>aws-architect-associate-4.pptx</code>	7	Expands
<code>aws-architect-associate-5.pptx</code>	8	Expands
<code>aws-architect-associate-6.pptx</code>	8	Latest and most complete — marked "V07 – 2018/11/01"

Deck 6 is a superset of the others and is the only one worth keeping as a source. The last three slides of deck 6 are empty.

The headline finding

These notes were written for **SAA-C01**, the exam version retired in 2020. The exam is now **SAA-C03**. The structure of your notes is good — compact, one box per service, which is exactly the right shape for memorizing — but a meaningful number of the facts have since changed, and several large topic areas that SAA-C03 tests heavily are not in the decks at all.

Read the two sections below before you drill anything. Unlearning a wrong number is much harder than learning a new one.

Facts in the decks that are now wrong

What the decks say	What is correct now	Why it matters
"5gb per object" (decks 1–5)	Max object size is 5 TB. 5 GB is the single-PUT limit.	Deck 6 fixed this to 5 TB. The earlier decks will cost you a mark.
"Lambda max execution time = 5 minutes"	15 minutes (900 seconds).	Deck 6 lists both the old 5-minute line and the correct 900 seconds, so it contradicts itself. Delete the 5-minute line.
"Lambda memory 128 MB to 3008 MB, in 64 MB increments"	128 MB to 10,240 MB, in 1 MB increments.	Changed in 2020.
"Lambda ephemeral disk 512 MB"	512 MB default, configurable up to 10,240 MB.	Changed in 2022.
"Max auto scaling group size is 20"	There is no hard limit of 20 — it is a soft quota you can raise.	This was never right and would lead you to a wrong answer on a scaling question.

What the decks say	What is correct now	Why it matters
"Data in Amazon S3 is not encrypted by default"	SSE-S3 is applied to all new objects automatically since January 2023.	A current exam answer saying "enable default encryption" may now be phrased differently.
"RRS — reduced redundancy storage 99.99%"	Reduced Redundancy Storage is deprecated. AWS no longer recommends it and it is not a valid exam answer.	Drop it entirely.
"Glacier: low cost storage 50tb"	There is no 50 TB figure for Glacier. Glacier is now three storage classes: Instant Retrieval, Flexible Retrieval, Deep Archive.	The whole Glacier box needs rewriting — see 04-storage.md .
"Expedited 1–5 min, standard 5–12 hours"	Expedited 1–5 minutes, Standard 3–5 hours, Bulk 5–12 hours.	Your notes merged Standard and Bulk and dropped Bulk. Three tiers, not two.
"CloudWatch retention period 14 days to 15 months" (deck 1) and "15 days to 15 months" (deck 6)	Retention is tiered by resolution : 1-second data 3 hours, 1-minute 15 days, 5-minute 63 days, 1-hour 455 days.	Neither version of the note is usable.
Extensive EC2-Classic detail	EC2-Classic was fully retired in August 2022.	Roughly a third of the EC2 box in every deck is about a platform that no longer exists. Delete it and reclaim the space.
"Classic ELB can scale proxy servers and backend instances"	Classic Load Balancer is legacy . The current three are ALB (layer 7), NLB (layer 4), Gateway Load Balancer (layer 3) .	Deck 6 added ALB, which is good, but still gives Classic top billing and omits GWLB.
"Multi-AZ deployments (multiregion is not available)"	Cross-region is available — RDS cross-region read replicas, and Aurora Global Database .	This would make you reject the right answer on a DR question.
"Aurora — 64 TB"	Aurora clusters now grow to 128 TiB. 64 TiB is the figure for the other RDS engines.	Two different numbers that your notes merged into one.
"SNS — note that FIFO queues are not currently supported"	SNS FIFO topics exist (since 2020) and deliver to SQS FIFO queues.	Out of date.
"Auto scaling: to use a new AMI create a new launch configuration template. One config per auto scaling group"	Launch configurations are legacy. Use a launch template , which is versioned, so you create a new version rather than a whole new object.	The mechanism in the note is the old one.
"WLM — can run multi-region or multi-az, for disaster recovery"	Workload Management is only about query queues and concurrency in Redshift. It has nothing to do with multi-region or DR.	Two unrelated ideas have been merged into one bullet.
"Kinesis Data Analytics"	Renamed Amazon Managed Service for Apache Flink .	The old name may still appear, but know both.

What the decks say	What is correct now	Why it matters
Origin Access Identity (implied by the CloudFront box)	Origin Access Control (OAC) is the current mechanism.	OAI still works but OAC is the exam answer.
"Spot block / fixed duration Spot"	Retired. Never the right answer.	Not in your decks explicitly, but worth knowing it is gone.

Structural problems in the decks

Problem	Where	Fix
The word "Redshift" is stranded inside the NACL text box	Decks 1 and 6, the NACL box reads "...CIDR — The /32 denotes one IP address... Redshift / NACL: Network ACL"	A copy-and-paste slip. It makes the NACL box confusing to revise from.
A VPN fact filed under IAM	Deck 6, the IAM box ends with "instances that you launch into a VPC can't communicate with your own network... attach a virtual private gateway..."	That belongs in the VPC box, not IAM.
Lambda appears twice on the same slide	Deck 1, slide 2 has two separate Lambda boxes	Merge them.
AMI appears twice	Deck 1, slide 2 has "AMI: Amazon Managed Images" twice with identical text	Merge them.
"Technical Tags, Automation, Business, Security" is attached to the CloudFront box	Decks 1 and 6	Those are the four tag categories . They belong only in the Tags box, where they also correctly appear.
AMI is expanded wrongly	All decks say "Amazon Managed Images"	It is Amazon Machine Image .
Typos that will trip you when revising aloud	"containers", "eventsi", "scriptf", "oosely coupled", "messages.o", "Datawarehouse"	Minor, but worth cleaning if you keep editing the decks.

What SAA-C03 tests that the decks do not cover at all

This is the bigger gap. Each of these has appeared on recent exams and none of it is in your notes.

■ Networking

- **Transit Gateway** — the transitive hub, and the answer whenever "many VPCs" appears. Peering is in your notes; Transit Gateway is not.
- **AWS PrivateLink / endpoint services** — exposing your own service to another VPC.
- **Global Accelerator** — static anycast IPs, TCP and UDP, and how it differs from CloudFront.
- **Gateway Load Balancer** — layer 3, for third-party appliances.
- **Route 53 Resolver** inbound and outbound endpoints for hybrid DNS.
- **IP-based routing policy** — your notes list seven Route 53 policies; there are now eight.
- **Egress-only internet gateway** for IPv6.

■ Security — this is the 30% domain, the largest on the exam

- **AWS Organizations and Service Control Policies**, including the heavily tested fact that **an SCP never restricts the management account**.
- **Control Tower** and **permission boundaries**.
- **Secrets Manager** versus **Systems Manager Parameter Store**.
- **GuardDuty, Inspector, Macie, Security Hub, Detective** — and the distinction between them, which gets tested directly.
- **AWS WAF** and **Shield Standard versus Shield Advanced**.
- **Firewall Manager** and **Network Firewall**.
- **ACM**, and the rule that a CloudFront certificate must live in **us-east-1**.
- **IAM Identity Center**, which replaced AWS SSO.
- The **policy evaluation order**: explicit deny, then allow, then implicit deny.
- **Envelope encryption** and the 4 KB direct-encryption limit on KMS.

■ Storage

- **FSx** — all four flavours. Your notes have EFS but no FSx at all, and "SMB or Windows file share" is a guaranteed exam phrase.
- **S3 Object Lock, Block Public Access, Access Points, Storage Lens, Batch Operations, S3 Select**.
- The **current storage class line-up**, including Intelligent-Tiering, Glacier Instant Retrieval, and Glacier Deep Archive.
- **Same-Region Replication** alongside Cross-Region Replication.
- **gp3**, which is now the default general-purpose volume and is cheaper and faster than gp2.
- **AWS Backup, Data Lifecycle Manager, DataSync, Transfer Family, Snowcone, Snowmobile, Tape Gateway**.

■ Databases

- **Aurora Serverless v2, Aurora Global Database, Backtrack, fast cloning**.
- **DynamoDB** depth: **Global Tables, TTL, transactions, point-in-time recovery, on-demand versus provisioned capacity, LSI versus GSI**, and partition key design. Your notes have four lines on DynamoDB and the exam asks far more.
- **ElastiCache Redis versus Memcached** — a direct comparison question that appears often.
- **Athena, Glue, Lake Formation, OpenSearch, Neptune, Timestream, QLDB, DocumentDB, Keyspaces, MemoryDB**.
- **DMS and the Schema Conversion Tool** for migration questions.

■ Application integration

- **EventBridge** — now a very common right answer, and not in your notes.
- **Step Functions** — your notes have SWF, which is the legacy service. Step Functions is what the exam wants.
- **SQS FIFO queues, dead-letter queues, long polling, visibility timeout tuning**. Your notes cover standard queues only.
- **Amazon MQ** for lift-and-shift of AMQP, MQTT, and JMS applications.
- **API Gateway** — your notes only say "can use API gateway" under Lambda.

■ Operations and cost

- **AWS Config** — the compliance leg of the CloudWatch / CloudTrail / Config trio. Your notes have the other two.
 - **Systems Manager**, especially **Session Manager**, which is the modern answer to "shell access without opening port 22."
 - **Compute Optimizer, Cost Explorer, Budgets, Cost Anomaly Detection, Savings Plans**.
 - **CloudFormation** depth — change sets, StackSets, drift detection, DeletionPolicy. Your notes have two lines.
 - **Well-Architected Framework** and the **disaster recovery strategies**. These frame a large share of SAA-C03 questions and are entirely absent.
 - **Outposts, Local Zones, Wavelength**.
-

What the decks get right and should be kept

Credit where it is due — these are accurate, well-chosen, and exactly the sort of thing the exam asks:

- The **security group versus NACL** comparison, including "security groups control ports on instances, NACLs control which networks reach the subnet." That is a good, memorable framing.
- The **five things a default VPC gives you**. Complete and correct.
- The **Route 53 routing policy** descriptions. Accurate, and only missing IP-based.
- The **Trusted Advisor CPFSS mnemonic**. Still correct, still useful. Keep it.
- The **S3 encryption breakdown** of SSE-S3, SSE-C, and SSE-KMS — correctly describing who holds which key. This is a common point of confusion and your notes have it right.
- **Multi-AZ for durability, read replicas for scalability**, which deck 6 states explicitly in the Terms box. That is the single most-tested database distinction and you have it.
- The **EC2 stop behaviour** list — EBS persists, instance store and RAM are lost, the instance usually moves to a new host. Correct and well worth keeping.
- The **default SSH user names** per AMI. Still accurate and still asked.
- The **CloudWatch default metrics** note that memory and disk need a custom script or agent. Correct, and a classic trap.
- The **volume status check** values of ok, impaired, and insufficient-data, every five minutes.
- The **Volume Gateway cached versus stored** distinction, added in deck 6. Correct.
- **CIDR /32** is one address and **/0** is the whole network.
- **SQS at-least-once delivery, order not preserved, 256 KB maximum** — correct for standard queues.

Recommended next steps

1. **Retire decks 1 through 5**. Keep only deck 6 as the historical source. Everything in 1–5 is either duplicated in 6 or superseded by it.
2. **Use the markdown notes in this folder as the working set**. They carry forward everything in the "keep" list above, correct everything in the "now wrong" table, and fill the gaps.
3. **If you want to keep a one-page visual**, rebuild it from **10-decision-triggers.md** rather than from the old slides. The trigger table is the part that earns marks on a scenario exam.
4. **Treat the .vce and exam-dump files in that folder with caution**. They are from 2017 to 2019, they target the retired SAA-C01, and dump questions are often wrong even for their own era. They are useful only for getting used to the question *format*, not for the answers. Use the official AWS practice question set and AWS Skill Builder for real practice.